

THE COGNITIVE TURN IN BEHAVIORAL ECONOMICS^{*}

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Abstract

This review organizes the vibrant recent literature on the cognitive foundations of economic decision-making, with a focus on empirical and experimental work. I review and interpret the evidence through the lens of uncertainty and information processing, distinguishing three interrelated stages of decision making: (i) the *acquisition* of information through attention and memory; (ii) the *interpretation* of information through mental models; and (iii) the *aggregation* of information into decisions under decision uncertainty and complexity. Reflecting its emphasis on information, research on cognitive foundations often studies beliefs, including in decisions traditionally viewed as reflecting non-standard preferences. This work has proved productive both by explaining and unifying several behavioral classics that were previously studied in isolation, and by generating new insights and regularities beyond the traditional scope of behavioral economics. The cognitive turn has reinvigorated upstream exchange with cognitive psychology and has begun to influence downstream applied research in fields such as finance, labor, development and macro.

Keywords: Cognition, attention, memory, complexity, uncertainty, information, mental models

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1 The Cognitive Turn

Suppose you're deliberating about your equity share—what fraction of your wealth to invest in stocks. Which variables and considerations do you pay attention to? The short-run financial needs you have because of your kids? That stock market crash from a few years ago? After you've converged on a set of variables, you realize that for some of them you don't remember all of the details. What was the average return of this asset class in the past again? How much did you spend on your kids each month? You also notice that for some variables you don't even know in which direction or how much they matter. How does the inflation rate affect the long-run tradeoff between stocks and bonds? What about AI? Finally, how do you deal with the complexity of aggregating all of this information into a decision? If the expected return of an ETF is 7% with a standard deviation of 20%, then is your expected-utility maximizing equity share 40% or 55%?

This example illustrates the importance of uncertainty, beliefs and different types of information processing in economic decision making. First, information is not always immediately available—instead, people need to *acquire information through attention and memory*. However, some variables may be more attention-grabbing or more memorable than others. Second, people need to *interpret this information through mental models*, by assessing how different variables affect outcomes. However, people's subjective models may be incomplete or misspecified. Third, even once potential outcomes are attended to, remembered and interpreted, people need to *aggregate this information into their best possible decision*. However, aggregating the primitives of a problem—including both external circumstances and one's preferences and beliefs—may be non-trivial for myriad reasons, including preference uncertainty, problem complexity, or the cognitive difficulty of maximizing.

Much recent research in behavioral economics (aka Psychology and Economics) studies these cognitive foundations of economic decision making: the processes of attending to, remembering, interpreting and aggregating information to reduce uncertainty and make decisions. Behavioral economics has traditionally focused on the 'high-level' constructs¹ of beliefs and preferences. The cognitive literature, instead, studies 'lower-level' cognitive processes—e.g., attention, memory, mental models, cognitive noise—that underlie observed beliefs and decisions. That said, the recent literature differs in critical ways from the earlier neuroeconomics wave summarized by Camerer (2013). In the classic taxonomy of the cognitive scientist Marr (1982), the recent literature largely studies the *algorithmic* level: which algorithms people use (e.g., similarity-based recall). Neuroeconomics, on the other hand, typically studies the most granular level of cognition, which describes how the brain *implements* an algorithm (e.g., how similarity-based recall is implemented in neural circuits).

To assess recent publication trends, I used AI² to classify all behavioral economics papers

¹I am grateful to Nick Barberis for suggesting this terminology.

²The downside of relying on AI for classification is that sometimes papers will be included in categories

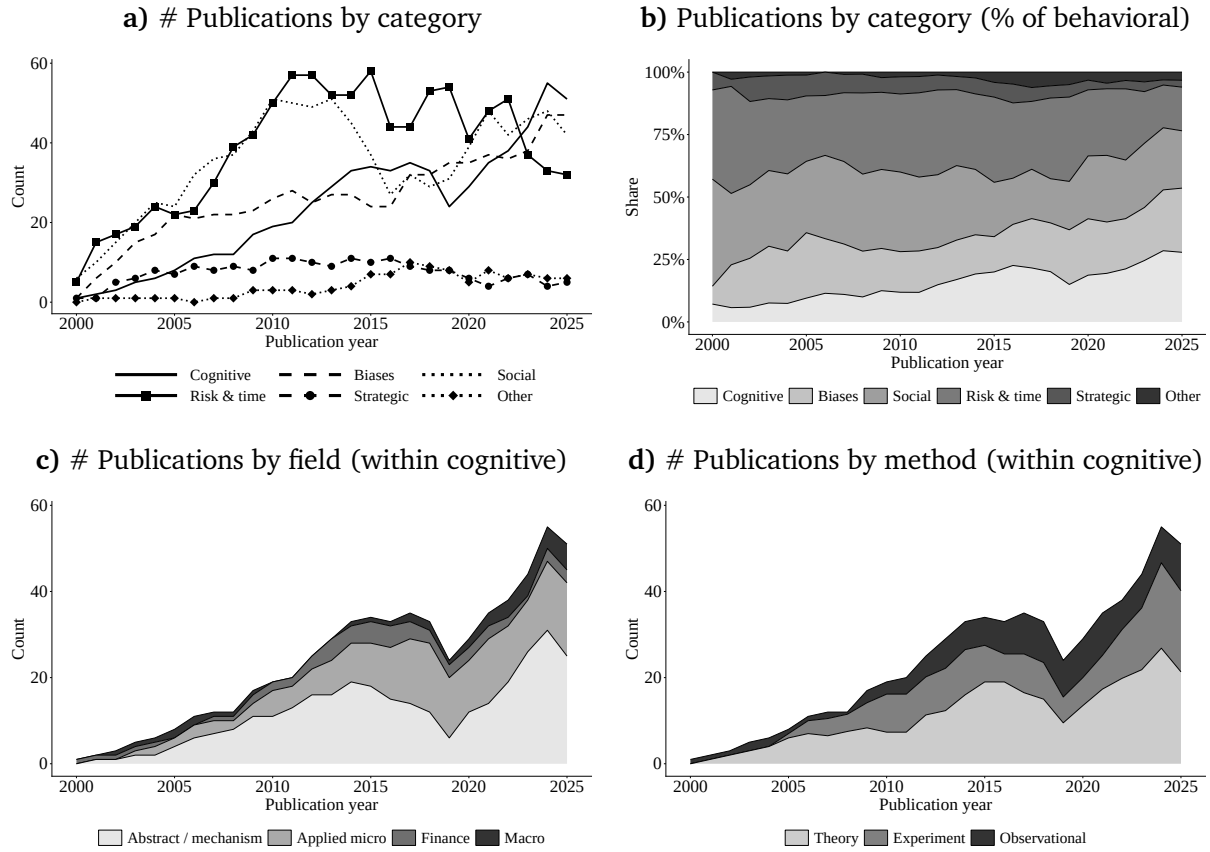


Figure 1: Composition of behavioral economics papers in ‘Top 5’ journals over time (three-year rolling averages). Top left shows counts of publications by category. Top right shows shares as a fraction of all behavioral papers. The bottom panels show distributions across fields / methods. Classifications performed by Claude Opus 4.6 based on paper’s title and abstract, see Appendix.

that were published in the profession’s ‘top 5’ journals between 2000 and 2025 into six categories based on their titles and abstracts: ‘cognitive’ topics (attention, memory, limited cognition, complexity, decision uncertainty, mental models), heuristics and biases, risk and time preferences, social behavior, strategic interactions, and others.

Figure 1 provides an overview of the publication trends. The overall picture is clear: research on cognitive topics accelerated along with work on biases and heuristics (thematically close to cognitive work). This came mostly at the expense of work on risk and time preferences, strategic thinking, and social motivations. Overall, the market share of cognitive work increased from 9.2% in the 2000-2005 period to 28.6% in the 2020-2025 period (a 19.4 percentage point increase / +211.1% relative change). This shift is not restricted to theory or abstract mechanism work but also partly driven by work in applied micro. These trends almost certainly underestimate the true magnitude of the cognitive turn because many of the papers that are discussed in this review are currently in the publication process. This review synthesizes and organizes this rapidly-growing body of work, focusing on the empirical and experimental literature.

even though they shouldn’t and that sometimes papers will be missing even though they should be included. The upside is that my hands are largely tied (up to the prompt reproduced in the Appendix).

Objectives and characteristics. The empirical literature on cognitive foundations has three primary objectives: partial unification, nuanced predictions, and an expanded scope of behavioral economics. Why should researchers outside of this literature be concerned with these objectives? Applied economists and theorists sometimes appear reluctant to leverage insights from behavioral economics for two interrelated reasons: a large number of concepts to choose from and instability. The worry appears to be that it isn't always clear which of the many departures from the standard model should be most important in which application and—maybe even more importantly—why any given behavioral phenomenon is found in some contexts but not in others. The objectives of the cognitive literature revolve around these concerns.

The first objective of the literature is to identify common cognitive mechanisms to bring together multiple behavioral economics phenomena (e.g., Fudenberg, 2006; Spiegel, 2019). For example, we often employ different approaches for understanding choice under risk, intertemporal choice and consumer choice, even when the empirical regularities and psychological principles may be very similar to each other. One hope in the literature is that an improved empirical understanding of cognitive foundations will facilitate a reduction in the number of behavioral concepts economists need to entertain. In this regard, an overly simplistic but perhaps helpful summary is that upon observing a non-standard choice pattern, the first reaction that motivates much research on cognitive foundations is to ask: “Which cognitive pattern generates this choice behavior, and how does it resemble regularities we’ve seen in other domains?”, rather than, for example: “Which utility function rationalizes this choice?” Still, the hope that *some* (or perhaps even *many*) empirical regularities may reflect common cognitive mechanisms does not mean that *all* regularities do. A ‘cognitive’ perspective does not deny the existence of non-standard preferences, but it shifts the focus to what people attend to, remember, infer, or are willing and able to aggregate.

The second objective is to facilitate more nuanced predictions. As noted, a recurring theme in studies of decision making is instability and context dependence. To take just one example, consider prospect theory, often taken to be the most successful behavioral model. A growing literature shows that its estimated parameters systematically depend on the elicitation technique used (e.g., Imai et al., 2025; Bouchouicha et al., 2024), vary with the relative salience of different payoffs (Bordalo et al., 2012; Frydman and Mormann, 2018; Bohren et al., 2024), and strongly correlate with proxies for cognitive uncertainty, complexity and dominance violations (e.g., Enke and Graeber, 2023; Oprea, 2024b; Qiu et al., 2026). One objective of work on cognitive foundations is to be able to better predict when and where behavioral anomalies occur.

A final objective of the literature is to expand the scope of behavioral economics. While we have made tremendous progress in incorporating some parts of psychology into economics, there are also some relatively under-utilized parts that are not obviously less important for economists. This includes several lines of work on information processing, such

as on attention, memory and exemplars, mental models, resource-rational cognition, the Bayesian brain hypothesis and decision confidence. This expansion in the psychology used will plausibly enable behavioral economists also to look at new economic applications.

Caveats. Two caveats are in order. First, this review is more of a progress report than a post-hoc synthesis. The cognitive turn is well underway but far from complete. Nonetheless, we have arrived at a juncture where reviewing and tying together different lines of work appears productive, both to synthesize and to highlight open questions.

Second, I'm aware of two main criticisms of the literature. The first is that work on cognitive foundations is sometimes one additional level removed from the decisions and market outcomes that are of central concern to economists. Proponents of work on cognitive foundations respond by highlighting the two potential upsides of the approach. First, that some reduction in tractability may be justified in exchange for greater parsimony in the number of concepts employed. Second, that understanding cognitive foundations does not hinder but instead enables applied economic analysis by highlighting the context-dependent nature of behavioral phenomena.

A second criticism is that the cognitive turn succumbs to a 'proliferation sin' akin to behavioral economics as a whole, by proposing an ever-expanding list of cognitive foundations. To address this concern, one of the main objectives of this review is to synthesize and highlight connections to make it transparent that the empirical results that have accumulated exhibit a substantial amount of structure. Because individual papers naturally emphasize differences with other papers to delineate their contributions, I focus on highlighting commonalities across empirical settings and decision domains.

Overview of broad lines of inquiry. It is common for behavioral economics reviews to be organized by decision domains. I deviate from this convention to highlight that, at least sometimes, the same cognitive principles underlie behavior across different domains. For instance, selective attention, memory and decision uncertainty matter not only for choice under risk, they also matter for labor supply and intertemporal or social decisions.

I illustrate concepts and fix terminology by means of a few simple equations. The objective is not to showcase a unifying model but, rather, to help orient the discussion and locate broad lines of inquiry. Table 1 provides an overview of the associated lines of work and some of their key empirical regularities.

Suppose a decision maker (DM) faces a vector of fundamentals (such as wages), $\theta = (\theta_1, \dots, \theta_J)$, and evaluates the utility of taking action $a \in \mathbb{A}$. Each potential action produces outcomes in separate problem dimensions (or states, or dates), $y(a, \theta) = (y_1^a, \dots, y_I^a)$. The outcomes are determined according to $y_i = m_i(a, \theta)$, where $m_i(\cdot)$ denotes the true model that maps the action and fundamentals into a dimension-specific outcome. For instance, θ_1 may be an hourly wage and a labor supply, in which case $y_1 = m_1(\theta_1, a) = \theta_1 a$ is total

earnings. Outcomes produce utility according to $U(a, \theta) = \sum v_i(y_i^a)$. This additively separable structure captures many contexts, including multi-attribute choice, choice under risk ($v_i = p_i u(y_i^a)$), intertemporal choice ($v_i = D(i)u(y_i^a)$), social preferences, and others. We then have that the DM’s problem is given by

$$\max_{a \in \mathbb{A}} U(a, \theta) = \max_{a \in \mathbb{A}} \underbrace{\sum v_i \left(\underbrace{m_i(\underbrace{\theta, a}_{\substack{\text{Acquisition} \\ \text{(attention, memory)}}})}_{\substack{\text{Interpretation} \\ \text{(mental model)}}} \right)}_{\text{Aggregation}}, \quad (1)$$

with solution $a^*(\theta) \in \underset{a \in \mathbb{A}}{\operatorname{argmax}} U(a, \theta)$. In this benchmark model, the decision maker (i) knows all relevant fundamentals and options; (ii) knows the correct model; and (iii) knows both how to aggregate problem features across dimensions and how to maximize. In practice, people may face uncertainty along any of these dimensions. Decision making thus requires the implementation of cognitive operations related to the acquisition, interpretation and aggregation of information (I collectively refer to these as ‘information processing’). While it isn’t always possible to unambiguously place a paper or empirical regularity into one of these ‘buckets’, we can nonetheless think of these three stages of decision making as motivating three interrelated literatures.³

First, the DM may need to acquire information about the fundamentals, θ , either by attending to them or by retrieving them from memory. For instance, the DM may not pay attention to or retrieve what the current inflation rate is. Throughout this paper, I loosely denote beliefs about quantity x by $\hat{x}$. Then, the literature proposes that the DM solves

$$\max_{a \in \hat{\mathbb{A}}} \hat{U}(a, \theta) = \sum \hat{w}_i(\cdot) v_i(m_i(\hat{\theta}, a)),$$

where $\hat{\mathbb{A}}$ is a consideration set, $\hat{\theta}$ is the perceived fundamental and $\hat{w}_i(\cdot)$ a subjective weighting function that reflects attentional or memory-based factors. This work, covered in Section 2, has highlighted the importance of both goal-driven processes (paying attention to what’s believed to be important) and stimulus-driven processes, such as low-level attentional mechanisms or cue-based recall.

Second, and related, the DM may not perfectly know the true model $m(\theta, a)$ that determines how fundamentals and actions map into outcomes. For instance, the DM may not know how the inflation rate affects GDP growth. Instead, the DM needs to interpret the data through the lens of his subjective mental model or problem solving approach:

³In a different terminology, ‘representations’ capture which features people take into account and how they are thought to matter, while ‘computations’ refer to the process of using some cognitive procedure—not necessarily an explicit mathematical one—to aggregate information into a decision. Ba et al. (2025) document the large gain in explanatory power that is achieved by studying these different stages simultaneously.

Table 1: Information processing in economic decision making

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graph LR
    A["I. ACQUISITION  
perceive  $\hat{\theta}$ "] --> B["II. INTERPRETATION  
form  $\hat{m}(\hat{\theta})$ "]
    B --> C["III. AGGREGATION  
choose  $\hat{a}^*(\hat{m}(\hat{\theta}))$ "]
    subgraph "mental representations"
        A
        B
    end
    subgraph "computations"
        B
        C
    end

```

Stage	Sec.	Literature	Empirical regularity	Examples
Information acquisition	2.1	Rational inattention	Focus on important	Stake size effects
	2.2	Salience, focusing	Focus on what stands out	Exaggerate large / diagnostic differences, skewness preferences
	2.3	Similarity-based memory	Recall what gets cued	Overreaction to news, stereotyping
Information interpretation	3.1	Misspecified mental models	Neglect relationships, variables or contingencies	Failure of contingent reasoning, correlation neglect, narrow bracketing
	3.2	Categorizations, analogies	Solve a related problem	Inference-forecast gap, similarity-weighted decisions
Information aggregation	4.2	Behavioral attenuation	Insensitivity to fundamentals	Probability weighting, hyperbolic discounting, attenuated equity shares
	4.3	Comparative thinking	Reference effects	Contrast and assimilation effects
	4.4	Caution, compl. aversion	Discount or avoid	Undervalue complex bets
n/a	4.1	Preference uncertainty and	Decision uncertainty	Cognitive uncertainty, deliberate randomization, choice revisions
	5	complexity	Complexity effects	Measure complexity costs, procedures; predict behavior with complexity

$$\max_{a \in \mathbb{A}} \hat{U}(a, \theta) = \max_{a \in \mathbb{A}} \sum v_i(\hat{m}_i(\theta, a)).$$

These subjective mental models may be misspecified (work on incorrect mental models), or the DM may try to solve a related problem instead of the true one (categorizations and analogies). This work is discussed in Section 3.

Finally, the DM may not perfectly know how to properly aggregate the fundamentals into an overall evaluation, $U(\cdot)$, or optimal decision, $a^*(\cdot)$. To illustrate, this literature posits that the DM's perceived optimal action is determined according to

$$\widehat{\max}_{a \in \mathbb{A}} \sum \hat{v}_i(y_i^a),$$

where $\widehat{\max}$ represents some less-than-fully-maximizing decision process. Empirical work in this vein, discussed in Section 4, often emphasizes that people rely on cognitive strategies that reduce the need to engage in a precise determination of value, such as noisy approximations, comparing with reference points, and shying away from difficult-to-assess options.

Finally, the framework laid out above is silent on what determines the magnitude of

information-processing requirements imposed by a decision problem. Research on the determinants of complexity, discussed in Section 5, can often be understood as studying this.

Section 6 concludes by integrating work on the different cognitive foundations to assess what we know about some behavioral economics ‘classics.’

Scope, objectives and interpretational lens of review. Given that some of the topics covered in this review have received attention in surveys of the theory literature (Gabaix, 2019; Woodford, 2020; Bordalo et al., 2022; Maćkowiak et al., 2023; de Clippel and Rozen, 2023; Vieider, 2025), I here focus on empirical and experimental work. I restrict attention to papers that—either in terms of their research hypothesis or the data collected—study a cognitive mechanism such as attention, memory, decision uncertainty or mental models.

Related to this paper are recent reviews on inattention (Handel and Schwartzstein, 2018; Loewenstein and Wojtowicz, 2025), contingent reasoning (Niederle and Vespa, 2023), experience effects (Malmendier and Wachter, 2024) and complexity (Oprea, 2024a).⁴

Relative to these contributions, this review has two main objectives. First, I aim to synthesize the results from different literatures through the interpretational lens of uncertainty and information processing – how people reduce uncertainty by acquiring and interpreting information through attention, memory and mental models; how decision uncertainty is visible in choices; how environmental cues provide information about reasonable courses of action; and how quantifiable features of decision problems allow us to predict the magnitude of information-processing requirements. Because this uncertainty and information lens suggests that multiple lines of work on cognitive foundations are linked, I will make the case that the literature contains an encouraging amount of structure.

My second objective is to provide an analytic guide to the literature, by identifying and organizing the techniques researchers have used to study cognitive foundations, and the (partly) converging evidence this has produced for understanding choice and beliefs.

2 Information Acquisition

Trivially, if we did not pay attention to and remember anything, we would exhibit maximal uncertainty about what to do. The main goal of the literature on information acquisition is to isolate predictable drivers of what gets attended to or remembered. Attention determines what gets encoded; memory determines what gets retrieved. While these form a sequential pipeline, I here discuss them partly in parallel.

⁴The intellectual origins of the ‘cognitive’ literature trace back to the bounded rationality movement (Simon, 1956, 1982) and the “system 1 vs. system 2” literature (Kahneman, 2011). Various additional contributions in the 1990s and early 2000s paved the way for the research that is taking place today, such as work on cues (Laibson, 2001; Mullainathan, 2002a; Bernheim and Rangel, 2004), analogies and categorization (Gilboa and Schmeidler, 1995; Mullainathan, 2002b), rational inattention (Sims, 1998, 2003), process tracing (Johnson et al., 2002; Camerer et al., 2005), choice uncertainty (Butler and Loomes, 1988; Ariely et al., 2003), and narrow bracketing and diminishing sensitivity (Prelec and Loewenstein, 1991; Read et al., 1999).

Throughout this paper, I use simple reduced-form equations to organize the evidence, spell out some of the key comparative statics predictions studied in the literature, and discuss some of the main empirical measurement tools and causal identification techniques. These equations will always be overly simplistic relative to the associated theory literature.

2.1 Goal-Directed Attention and Memory

Organizing equation and empirical tools. Work on goal-directed attention and memory can heuristically be represented by⁵

$$\hat{U}(a, \theta) = \sum v_i(m_i(\hat{\theta}, a)) \quad (2)$$

$$\hat{\theta}_j = \lambda_j \theta_j + (1 - \lambda_j) d_j, \quad \lambda_j = \beta \pi_j + (1 - \beta) \alpha. \quad (3)$$

Here, the DM only attends to or remembers a subset of the available information and accounts for this by partially relying on some prior or default value. Crucially, the amount of information acquisition, captured by $\lambda_j \in [0, 1]$, potentially increases in the importance of the variable for the decision, here captured in a reduced-form way by $\pi_j \in [0, 1]$, where α is some baseline level of attentiveness.

The main comparative statics test conducted in this literature is whether higher incentives increase information acquisition (attention), and by how much. To study this question, the literature has relied on a broad set of contexts, including choice and visual perception experiments, field experiments and expectations data. Table 2 provides an overview of relevant papers.

Experimental evidence. Through the lens of equation (3), I highlight three recurring results in the experimental literature. First, consistent with the presence of imperfect information acquisition, the sensitivity of proxies for the perceived quantity $\hat{\theta}$ to variation in the objective quantity, θ , is consistently smaller than one, i.e., $\lambda < 1$. This shows that people are partially inattentive. Second, *qualitatively* consistent with a role for goal-driven or maximizing attention allocation, this degree of insensitivity generally decreases as the incentives increase, i.e., λ_j increases with π_j . Third, the *quantitative* responsiveness of attention to variation in incentives is often smaller than predicted, i.e., $\beta < 1$.

Gabaix et al. (2006) study information acquisition in an experiment with explicit financial or time costs. For instance, in one experiment, subjects choose between options involving several hidden payoffs and can pay a cost to reveal a payoff. Subjects respond to incentives in predicted ways, acquiring more information when the economic value of information is higher. At the same time, the slope of this relationship is smaller than predicted. Dean and Neligh (2023) report a conceptually similar result in a visual perception

⁵See, for example, Sims (2003), Caplin and Dean (2015), Maćkowiak et al. (2023), and Gabaix (2014).

Table 2: Evidence on responsiveness of attention to incentives

Decision	Information acquisition	Variation in incentives	Inattention? ($\lambda < 1$)	Incent. effect?	Magnitude as pred.? ($\beta = 1$)	References
Evidence from lab experiments						
Choose multiattribute good after sequent. info search	Reveal attribute values	Magnitude of values	Yes	Yes	No, smaller	Gabaix et al. (2006)
Psychometric task: estimate freq. of objects	Visual / response time	Monetary reward	Yes	Yes	Not analyzed	Caplin et al. (2020)
Estimate value of asset	Historical price path	Monetary reward	No (on avg.)	Yes	Not analyzed	Hartzmark et al. (2021)
Survey completion	WTP for reminder	Monetary reward	Yes	Yes	No, smaller	Bronchetti et al. (2023)
Accurately solving math problems	WTP for reducing difficulty	Monetary reward	Yes	Yes	No, smaller	Bronchetti et al. (2023)
Psychometric task: estimate freq. of objects	Visual (unobserved)	Monetary reward	Yes	Yes	No, smaller	Dean and Neigh (2023)
Evidence from field experiments and surveys						
Evaluate candidates for jobs / evaluate potential renters	Reveal characteristics	Cherry-picking vs. lemons-dropping market	Yes	Yes	Not analyzed	Bartoš et al. (2016)
Decide whether or not to purchase good in presence of sales tax	Attention to tax (unobserved)	Tax rate	Yes (on avg.)	Yes	Not analyzed	Taubinsky and Rees-Jones (2018) Morrison and Taubinsky (2023)
Completion of online coding course	Take-up of plan making tool	Monetary reward	Yes	Yes	No, smaller	Bronchetti et al. (2023)
Indicate house price expectations w/ accuracy incentives	Purchase information	Monetary reward	Yes	Yes	Not analyzed	Fuster et al. (2022)
Whether or not mention inflation in open-ended surveys about what's top of mind for HH and managers	Real life (self report)	Exposure to inflation	Yes	Yes	Not analyzed	Link et al. (2023)
Survey self-reports about whether know inflation rate	Real life (unobserved)	Inflation rate	Yes	Yes	Not analyzed	Bracha and Tang (2025)
Expectations update after inflation info provision	Real life (unobserved)	Inflation rate	Yes	Yes	Not analyzed	Weber et al. (2025)

experiment motivated by the rational inattention literature. They experimentally vary the incentives for correctly identifying which of two colors is more prevalent on a computer screen. They find that accuracy increases in the stake size as predicted, but that this relationship is substantially weaker than predicted by rational inattention models.

Bronchetti et al. (2023) provide an important illustration of under-responsiveness to variation in incentives by studying the demand for tools that reduce attention or memory costs across multiple lab and field experiments. For instance, in a field experiment, they measure demand for a plan-making tool that helps students on an online education platform to complete the required course modules. Bronchetti et al. (2023) experimentally vary the incentives to complete the course. This increases demand for the attention (plan making) tool but again the effect is smaller than predicted by optimizing attention allocation models.

Evidence from applied settings. A substantial body of work studies information acquisition in field settings, including in public finance and macroeconomics. This work can often be interpreted through the lens of both goal-driven attention and goal-driven memory. Especially when surveys measure people’s knowledge about variables such as inflation, the resulting patterns could reflect forgetting as much as inattention. In applied work, three empirical techniques have been deployed to measure information acquisition: (i) direct measures of information search, (ii) self-reported attention/knowledge, and (iii) inferring attention from the magnitude of belief updating after information provision.

In the first category, Bartoš et al. (2016) study discrimination through endogenous information search by tracking employers’ and landlords’ online information acquisition. The authors send email applications to employers and landlords and then track their information acquisition by recording how intensively they search the candidate’s website. When evaluators aim to identify the best applicants (such as in a ‘cherry-picking’ labor market), they search less about minority groups they expect to be lower quality. On the other hand, when evaluators aim to avoid the worst (such as in a ‘lemons-dropping’ rental market), they search more about the minority groups. These results show that costly, goal-directed attention can generate discriminatory outcomes through selective and asymmetric search.

In the second category, Link et al. (2023) use surveys with a household panel and a firm panel to study time variation in self reports of what is top of mind among managers and households. They identify three pieces of evidence that attention allocation responds to incentives. First, inflation is substantially more frequently mentioned as being top of mind in open-ended survey questions when inflation is higher. Second, exposure to energy prices is strongly correlated with inflation being top of mind for households and managers. Third, this latter effect is especially pronounced during times of high volatility. Similarly, Bracha and Tang (2025) show that the share of households who report not knowing the inflation rate decreases in inflation, again consistent with incentive effects.

Finally, in the third category, Weber et al. (2025) study the impact of information inter-

ventions across a large number of surveys on inflation expectations. They find smaller belief updates to inflation information when inflation is higher. This is consistent with an incentives effect and suggests higher baseline attention (and thus less ‘surprise’ from the information provision) in high-inflation periods.

2.2 Stimulus-Driven Attention: Contrast and Exaggerating Differences

Organizing equation and empirical tools. Work on stimulus-driven attention can be organized by variants of the equation⁶

$$\hat{U}(a, \theta) = \sum \hat{w}_i(y_i^a, y_i^{-a}) v_i(y_i^a), \quad (4)$$

where $\hat{w}_i(y_i^a, y_i^{-a}) = f(|y_i^a - y_i^{-a}|)$ is a context-dependence or salience weighting function that typically increases in the absolute distance between the outcome associated with the action under consideration and other relevant outcomes in the choice or comparison set. We can thus think of payoffs as having a direct effect on choices (payoffs affect utility) and an indirect information-acquisition effect (payoffs affect what stands out and thus change perceived utility).

This literature thus posits that information acquisition involves a *comparative* thought process according to which people acquire more information about (pay more attention to) an outcome when it is very different from other outcomes, the general idea being that changes, surprises and variability attract attention. For example, a large change in the state of the stock market attracts more attention than a small change. One potential interpretation of this is that dimensions in which outcomes exhibit more variation are generally more important for decisions (trivially, dimensions with no variation cancel out), and that people heuristically treat the presence of large variation as informative about which dimensions are important even when such inferences are incorrect.⁷

The main empirical signature of such attention allocation is a kernel-of-truth overreaction: people respond to real payoff or likelihood differences, but overweight dimensions or states where contrasts are large or surprising (e.g., Bordalo et al., 2016). Table 3 provides an overview of studies in the literature that have provided evidence for the broad idea that large or surprising differences attract attention and / or get overweighted in choice and belief formation. As an analytic guide, the table highlights the technique used by the respective researchers to induce variation in stimulus-driven attention. This work sometimes relies on process-tracing techniques such as eye tracking and mouse tracking, or on predicting visual salience using machine learning algorithms. To exogenously vary salience and attention, researchers rely on a combination of techniques, often manipulating the magnitude of dif-

⁶See, for example, Bordalo et al. (2012, 2013, 2016) and Kőszegi and Szeidl (2013).

⁷Below in Section 4.3 I will discuss a second type of comparative reasoning, in which comparisons help with (or provide information about) how to navigate tradeoffs or aggregate across dimensions.

ferences to study whether larger differences receive more weight.

Choice. I first summarize lab evidence, then discuss visual salience and field applications.

A first line of work focuses on decision making under risk, such as the Allais paradoxes or skewness preferences. One methodological idea in this literature is to manipulate the correlation structure among different assets. As noted above, the reason is that the idea of differences attracting attention envisions a *comparative* decision process, in which the weight attached to different states (or dimensions) depends on the magnitude of differences or surprises in that state. Thus, changing the correlation structure among assets while holding their marginal distributions constant affects the magnitude of differences, while holding the problem constant from many other perspectives. Along these lines, Bordalo et al. (2012) document that the common-ratio and common-consequence Allais paradoxes can both be switched on and off in systematic ways by changing the correlation structure among the lotteries. For instance, in the case of the common-consequence paradox, the payout states can be rearranged such that the common consequence is more or less salient. Using similar techniques on correlated state spaces, Dertwinkel-Kalt and Köster (2020, 2025) provide evidence on how stimulus-driven attention allocation drives skewness preferences.

In a particularly direct test of the idea that differences attract attention, several authors have varied the range of outcomes in a dimension by manipulating the location of a decoy good (Castillo, 2020; Andersson et al., 2021; Bondi et al., 2025). All of these papers document that expanding the range of attribute values in a dimension shifts choice towards the option that is better in that dimension, the underlying mechanism presumably being that large variation attracts attention. Bondi et al. (2025) document that this effect is substantially stronger when (i) the number of attributes is larger, and (ii) when decision makers are exogenously placed under higher cognitive load. This suggests that the idea of differences attracting attention is especially important in contexts in which the cognitive constraint for information acquisition binds. This is a recurring insight that I return to below.

Dertwinkel-Kalt et al. (2022) provide evidence for the idea that differences attract attention in the context of effort supply. Their experiment tests the idea that overweighting large differences implies a form of ‘concentration bias,’ according to which people prefer options that have one large advantage and many small (and thus less noticeable) disadvantages. In their experiment, exerting more effort has very dispersed costs (on each of 10 days), yet the benefits are concentrated (a higher one-time reward). They find that—relative to a baseline condition with balanced tradeoffs—subjects predominantly choose the option with the concentrated advantage, acting as if the large reward gets overweighted.

Beliefs. Research on stimulus-driven attention in choice and belief formation exhibit a close analogy. Both lines of work posit that (i) differences attract attention and (ii) that there is a kernel of truth, according to which the differences to which people respond are

Table 3: Evidence on stimulus-driven attention and exaggerating differences

Decision	Variation in salience induced by	Result	References
Evidence from choice experiments			
Binary lottery choice	Correlation structure among lotteries	Switch Allais paradox on / off	Bordalo et al. (2012); Frydman and Mormann (2018) Bruhin et al. (2022)
Binary lottery choice	Correlation structure among lotteries	Skewness pref. (overweight states with large differences)	Dertwinkel-Kalt and Köster (2020, 2025)
Trinary choice	Decoy manipulates attribute range	Overweight dimension with large variation	Castillo (2020); Andersson et al. (2021)
Binary effort choice	Effort cost concentrated vs. spread out over days	Overweight dates with large differences	Dertwinkel-Kalt et al. (2022)
Binary effort choice	Decoy manipulates attribute range	Overw. dimension w/ large variation more when more attributes	Bondi et al. (2025)
Strategic coordination	Visual salience in image	Choose salient location	Li and Camerer (2022)
Investment decision	Visual salience in price chart	Overweight high-salience returns	Bose et al. (2022)
Binary asset choice	Visual outlier in payoff chart	Overweight outlier	Bohren et al. (2024)
Evidence from experiments on beliefs and expectations			
Belief about state	Representativeness of signal for state	More overreaction when more states	Ba et al. (2025)
Belief about attribute frequency in group A	Attribute frequency in group B	Belief about frequency in A decreases in frequency in B	Bordalo et al. (2016)
Beliefs about women	Performance of men	Exaggerate true ability differences	Bordalo et al. (2019)
Belief about attribute	Average attribute in other group	Exaggerate true differences	Esponda et al. (2023)
Evidence from the field			
Asset holdings	Visual salience of capital gains	Stronger disposition effect	Frydman and Wang (2020)
Stock returns	Difference b/w asset returns and market returns	High salience-weighted past returns predict low returns	Cosemans and Frehen (2021)
Beliefs about likely job given college major	Representativeness of major for job	Beliefs exaggerate true differences; associations predict beliefs	Conlon and Patel (2022)

real, yet people overreact as a result of stimulus-driven attention allocation.

The basic intuition behind this line of work is a contrast effect. Suppose a decision maker assesses the frequency of an attribute in group A, while simultaneously being exposed to group B. Then, the idea of contrast (overreaction to larger differences) is that the estimate for group A decreases in the estimate for group B, giving rise to stereotyping (e.g., Bordalo et al., 2016). This evidence is particularly direct because, in clean experimental settings, groups A and B are independent, yet nonetheless the presence of B affects the perception of A. This evidence spans abstract lab settings (Esponda et al., 2023) and applied domains such as gender beliefs, where beliefs about women’s absolute ability vary as a function of men’s absolute ability in a contrast direction (Bordalo et al., 2019).

In the context of updating beliefs about an underlying state, one way to quantify the degree to which a feature of the problem attracts attention is through its representativeness (Kahneman and Tversky, 1972; Camerer, 1987; Bordalo et al., 2016). The idea is that when

a statistic is highly diagnostic of one state—i.e. if the difference in the likelihood of observing different signals is large for a given state—then the state attracts attention, in line with the contrast logic spelled out above. Various papers have provided evidence for this broad idea using a combination of data on beliefs and attention.

Similarly to Bondi et al. (2025), Ba et al. (2025) hypothesize that the attention-grabbing effect of large differences is more pronounced when the decision problem is more complex (involves a larger number of states). In belief updating experiments, they find that when the number of states is small, people are relatively insensitive to variation in the signal strength (foreshadowing the work on behavioral attenuation reviewed below). Yet as the number of states increases people appear to rely more and more on the attention-grabbing heuristic of focusing on states in which the difference of the likelihood of different signals is large. Crucially, Ba et al. (2025) use mouse-tracking techniques to show that subjects indeed pay substantially more attention to a state when it is representative, and that this attention allocation predicts overreaction in beliefs. Again, information acquisition thus introduces an additional 'indirect effect' of a normatively-relevant feature (the diagnosticity of a signal for different states), which biases the belief updating process.

The general idea that people's beliefs reflect the exaggeration of true differences has also been studied in field applications. For example, Conlon and Patel (2022) document that undergraduate students strongly stereotype the link between college major choice and subsequent occupational outcomes – they exaggerate the likelihood that enrolling in a major will lead them to take up a job that is most representative of that major. They complement evidence on these exaggerated belief differences with an implicit associations test, finding that people who more strongly associate occupations with the college majors for which they are most representative, have significantly more distorted beliefs. This suggests that low-level associations—based on exaggerating true differences—affect high-stakes ecological beliefs and major choice.

Visual salience. Most of the experimental literature on salience and focusing investigates the idea that context dependence is driven by payoffs or objective attribute features. A complementary line of work has highlighted the role of visual salience. Li and Camerer (2022) use machine learning algorithms that are trained on gaze data to predict visual salience. They document in a consumer choice experiment that participants are less likely to make choice mistakes if the higher-value option is also the visually salient one. In a related study, Bose et al. (2022) show that forecasts of autocorrelated processes based on standard price charts significantly depend on which part of the price chart is visually salient. For example, when positive past returns are visually salient, people overestimate future returns and invest more into the respective stock.

Related, Bohren et al. (2024) document in a lottery choice context that people strongly overweight visual outliers in a return chart. Their contribution is original in part because

it provides direct evidence on the underlying attention allocation. Using a combination of mouse tracking and eye tracking techniques, they find that outliers in a price chart indeed attract more visual attention, and that such selective attention is strongly predictive of choices. Bohren et al. (2024) emphasize that this stimulus-driven selective attention can produce decision anomalies that were previously ascribed to preferences (e.g., probability weighting), posing a challenge for choice-based preference identification and welfare analysis.

A small body of work studies visual attention in field contexts. Frydman and Wang (2020) leverage a natural experiment in which some traders are randomly exposed to a trading interface in which past capital gains are substantially more visually salient. They show that this produces an increase in the disposition effect, again consistent with the idea that visual salience shapes decision weights.

Work on economic salience (as driven by payoffs or returns) has a long tradition in finance. For instance, Barber and Odean (2008) show that retail investors are net buyers of stocks that have experienced abnormal returns and are thus attention grabbing. More generally, there is a large literature on inattention in finance (e.g., DellaVigna and Pollet, 2009), though this work is often more concerned with documenting inattention rather with identifying which features determine what people pay attention to.

Summary. Taken together, this body of work shows two main patterns. First, differences attract attention, as reflected in mouse- and eye-tracking studies. Second, this is linked to a contrast effect in beliefs and choices: true differences that should matter—between groups, options or states of the world—get exaggerated because the process of endogenous information acquisition further amplifies the effect of differences on choices and beliefs.

2.3 Stimulus-Driven Memory

Organizing equation and empirical tools. Work on stimulus-driven memory can be represented as positing that variables (or outcomes) are retrieved as⁸

$$\hat{\theta}_j = \omega_j \theta_j + (1 - \omega_j) d_j, \quad \omega_j = \frac{S(c_j, \theta_j)}{\sum_k S(c_j, \theta_k)}. \quad (5)$$

Here, recall ω_j increases in the similarity between the object to be retrieved and the present problem or context, c_j . This captures the idea of cue-dependent recall. Simultaneously, recall decreases in the similarity between the current cue and other items in the memory database, a process called interference (the denominator of ω_j). A main idea is that objectively irrelevant events or contextual features can shape beliefs because they change the similarity between the current context and the variable the decision maker is trying to retrieve. For example, when forming beliefs about the future evolution of the stock market, observing

⁸See, for example, Kahana (2012); Bordalo et al. (2020b, 2023).

Table 4: Evidence on stimulus-driven memory

Decision / belief	Variation in contextual feature / similarity	Result	Memory corr.	References
Evidence from experiments				
Estimate frequency of experienced objects	Interference from other objects	Higher estimate when less interference	Yes	Bordalo et al. (2023)
Estimate cyber attack risk	Priming with (dis)similar events	Estimate more strongly affected by similar events	Yes	Bordalo et al. (2025a)
WTP for / belief about value of asset based on signals	Cue similarity with past signals	Overreaction to news when stable link b/w signal and cues	Yes	Enke et al. (2024b)
Recall frequency of positive restaurant review	Review is story vs. statistic	Stories more memorable b/c more similar to prompt	Yes	Graeber et al. (2024)
Estimate frequency of random event	Cuing with example outcomes	Cues more similar to event increase belief in event	n/a	Conlon and Kwon (2025)
Estimate political partisan extremity	Similarity b/w positions and typical partisan positions	Predominantly remember extreme / stereotypical info	Yes	Cai (2022)
Evidence from the field				
Estimate future stock returns	Today's return	Higher remembered return when today's return higher	Yes	Jiang et al. (2025)
Inflation expectations	Negative income or health shocks	Forecasts / memories pessimistic after income / health shocks	Yes	Butera et al. (2024)
Inflation expectations	Similarity to past experiences	Stronger reliance on past experience when more similar to present	n/a	Gennaioli et al. (2024)
Inflation top of mind in open-ended survey	Similarity to past inflation experiences	More attention following shock experiences when prior shock experience	n/a	Link et al. (2023)
Consumer demand as fct. of ads	Ad similarity	Ads for similar products increase demand	n/a	Bordalo et al. (2025b)
Savings/ predict future expenses	Categories that cue expenses	Expenditure beliefs and savings increase when expenses cued	Yes	Augenblick et al. (2023)
School track rec./ student evaluations	Similarity b/w perf. and gender stereotype	Remember stereotype-consistent info → discrimination	Yes	Miserocchi (2023)

high returns today (perhaps in conjunction with vivid good-times stories) may selectively cue the recall of high past returns, and thus produce beliefs that are too optimistic.

Table 4 provides an overview of empirical studies, covering work on experiments, surveys and observational data. As an analytic guide to the literature, I highlight the measurement techniques deployed (recall, beliefs) and the variation in similarity or interference that is leveraged to study the outcome of interest.

Evidence from the lab. A frequently-used strategy to document the economic effects of similarity-based recall is to induce exogenous variation in contextual similarity, holding the objective informational content fixed.

Bordalo et al. (2025a) use variants of priming techniques to study the effect of cues on beliefs. They elicit people's beliefs about the risks from cyber attacks (among other things).

Before eliciting these beliefs, they cue people about similar or dissimilar risks, for instance by asking whether the respondent has experienced an ID theft (which respondents judge to be similar to a cyber attack), or by asking whether the respondent has experienced the loss of a loved one (judged as dissimilar to cyber attacks). They find that cuing similar events increases beliefs about the risks from cyber attacks.

Enke et al. (2024b) study the role of similarity-based recall in driving overreaction of expectations. Their main hypothesis is that when similar news items are consistently associated with similar contextual features (e.g. positive stock returns typically co-occur with good-times stories), then similarity-based recall will produce overreaction to information because the contextual feature will selectively cue the retrieval of similar past news items. In their experiment, subjects sequentially receive information about an asset's value, where each piece of news is communicated with a memorable story and an image, which serve as cues. When the same news items are consistently associated with the same cues, beliefs and measured recall strongly overreact to recent information. On the other hand, when there is no stable mapping between cues and news (such that associative recall is by design shut down), there is no overreaction.

Graeber et al. (2024) study the role of similarity-based recall for whether stories or statistics (data) play a more important role in affecting long-run beliefs. They show that people incorporate both quantitative data and qualitative stories into their beliefs at the encoding stage, but forgetting is substantially more pronounced for statistics than for stories. The authors provide evidence that this occurs because stories get more effectively cued and are thus more effectively retrieved. Intuitively, stories are usually more unique than numbers and thus subject to less interference, making them easier to remember.

Evidence from the field. A growing literature studies the role of similarity-based memory in the field, in contexts ranging from finance to macro to discrimination to development.

Jiang et al. (2025) document the relevance of associative recall by combining surveys of investors' memories with administrative trading data. The headline result is that when market returns are positive, people become more likely to remember episodes of rising markets from the past, a hallmark signature of associative recall. These investor memories of past returns, in turn, are strongly predictive of future return expectations and extrapolation, suggesting that current returns selectively cue past returns, which are then used to form predictions about future returns. This work is potentially related to the evidence on pronounced experience effects, according to which people's inflation beliefs and stock market behavior are in large part shaped by their own inflation and stock market experiences (Malmendier and Nagel, 2011, 2016).

In macro, a stream of papers has linked similarity-based memory to inflation expectations. Link et al. (2023) use open-ended survey questions to measure what is top of mind for households and managers. They find that respondents who experienced the oil crises

and associated inflation surges are more likely to have inflation top of mind. Crucially, this experience effect is more pronounced when current inflation is high, consistent with a cue-based recall mechanism. Gennaioli et al. (2024) provide evidence on how this mechanism affects expectations. By analyzing survey data, they find that past inflation experiences are more predictive of current forecasts when they are more similar to current conditions.

Related is also the study of Butera et al. (2024) on inflation expectations. Linking survey data on inflation forecasts with administrative data, they find that inflation forecasts and memories are both strongly affected by idiosyncratic income and health shocks. This suggests that negative shocks drive negative affect, which in turn cues bad memories.

Augenblick et al. (2023) study the importance of imperfect and associative recall in a development context, by analyzing the savings patterns of Zambian farmers. In surveys of the farmers, they document that people have a pronounced tendency to forget about upcoming expenditures that are small, irregular and stochastic, and as a result save too little. To improve recall and savings, the authors leverage the idea of associative recall by asking farmers to provide estimates of future expenditure in each of several categories. The idea is that the provision of expenditure categories cues the retrieval of associated expenditure events. The authors find that in this treatment condition people become better able to predict future expenses, which produces a large increase in savings.

Finally, a growing body of evidence identifies selective memory as a driver of stereotyping. The idea in this line of work is that information about an individual gets more easily retrieved when it is more representative of the group to which the individual belongs. This is found both in abstract lab experiments (Bordalo et al., 2021) and in more applied settings. For instance, Miserocchi (2023) studies the role of associative and representativeness-based recall for gender discrimination. In a lab-in-the-field experiment in Italy, teachers provide high school track recommendations for students and are asked to recall performance data on these individual students. Miserocchi (2023) finds that teachers disproportionately retrieve stereotype-consistent information ('girls are bad at math'). As a result, girls are less likely to be recommended for math-intensive tracks. In contrast, when memory constraints do not bind because teachers are reminded of each student's performance data, the gender gap in high school recommendations shrinks substantially.

3 Information Interpretation

Once information is acquired through attention and memory, it needs to be interpreted through the lens of some mental model. Perhaps inflation is high, but does that actually matter for whether I'm likely to lose my job? Perhaps all of my friends tell me that they like the new restaurant in town, but do these reports reflect independent information?

In contexts like these, the decision maker faces uncertainty about the true model. Thus, he needs to engage in two types of information processing. First, to process the available

information to determine what the right mental model $\hat{m}$ is (the focus of this section). Then, to aggregate the available information through the lens of his subjective mental model to arrive at a decision (the focus of the next section). Yet while superficially related, this section will highlight one key distinction between uncertainty about models and uncertainty about aggregation – uncertainty about models sometimes leads people to fully neglect a problem aspect and jump to others, while uncertainty about aggregation is usually more strongly associated with continuous shading.

3.1 Incomplete Representations

Empirical tools. Work on ‘incomplete representations’ studies settings in which people appear to entirely miss a variable or causal link that actually matters. For instance, if the true model is given by $m = \sum b_i \theta_i$, then the literature emphasizes that frequently $\hat{b}_i = 0$ for some dimension i .⁹ This pattern appears in four regularities: (i) system neglect (missing edges in the data-generating process), (ii) neglecting model uncertainty, (iii) neglecting statistics (discarding presented evidence because it’s deemed irrelevant), (iv) compartmentalization across problems, and (v) neglecting contingencies (missed states/conditionals).

Methodologically, this literature often proceeds by backing out people’s mental representations from elicited beliefs. The general idea is that an incomplete representation is indicated by people not responding at all to variation in the omitted factor. A recurring technique is to show that people’s representations (and resulting decisions) can be corrected by drawing their attention to the omitted factor, and by showing that incomplete representations are more likely to arise when the problem is more complex.

System neglect. There is much evidence in psychology and decision research to suggest that people tend to excessively focus on the salient, immediately visible ‘output’ (data) that a causal structure produces, rather than on the features of the underlying data-generating process (e.g., Massey and Wu, 2005; Fiedler and Juslin, 2006). The majority of experimental results in economics that have this structure are found in the domain of belief updating. Table 5 provides an overview. I restrict attention to papers that (i) document full neglect of some problem aspect and (ii) show links with either attention or complexity.¹⁰

A series of experimental papers shows that people often fully neglect ‘quirks’ in the data-generating process, treating the signals at face value, as if they are generated by an unbiased iid process. In this class of experiments, subjects are asked to form beliefs based on exogenously-provided signals. These signals are either correlated and partially redundant (Enke and Zimmermann, 2019), exhibit a selection problem (Enke, 2020), or reflect

⁹See, for example, Gabaix (2014) and Bordalo et al. (2025c).

¹⁰Potentially related to this evidence is work on neglect of indirect effects, most notably of equilibrium effects (Dal Bó et al., 2018; Nunnari et al., 2024; Dreyfuss, 2025). However, at this point I am not aware of direct evidence that links equilibrium neglect to attention and / or complexity. For instance, Abeler et al. (2025) document the role of complexity resulting from indirect effects in workers’ incentive contracts.

Table 5: Evidence on incomplete representations (‘full neglect’)

Phenomenon	Role of attention	Role of complexity	Awareness	References
Evidence from experiments				
Correlation neglect among signals	Nudge reduces neglect	Number of signals $\uparrow$ $\rightarrow$ neglect $\uparrow$	n/a	Enke and Zimmermann (2019)
Selection neglect among signals	Nudge reduces neglect	Comput. complexity $\uparrow$ $\rightarrow$ neglect $\uparrow$	Largely unaware	Enke (2020)
Neglect of alternative causes	Nudge reduces neglect	n/a	Largely unaware	Graeber (2023)
Base rate or signal neglect	Salience predicts neglect	n/a	n/a	Bordalo et al. (2026)
Base rate neglect	Ignoring data predicts neglect	Memory burden $\downarrow$ $\rightarrow$ neglect $\downarrow$	Largely unaware	Esponda et al. (2024)
Model uncertainty neglect	Attention on one model	Comput. complexity $\uparrow$ $\rightarrow$ inattention / neglect $\uparrow$	Largely unaware	Musolf and Zimmermann (2025)
Model uncertainty neglect	Attention on one model	n/a	n/a	Aina and Schneider (2025)
Attribute neglect	Nudge shifts neglect	n/a	n/a	Conlon (2025)
Evidence from the field				
Attribute neglect	Nudge reduces neglect	n/a	Largely unaware	Hanna et al. (2014)
Expected earnings reasoning for stock returns	No effect of nudge	n/a	n/a	Andre et al. (2023)

alternative causes (Graeber, 2023). Across these settings, bi-modal belief patterns obtain, according to which subjects act either Bayesian or as if they exhibit an incorrect mental model that fully equates the visible messages with the underlying signals. A recurring result in this literature is that drawing people’s attention to the previously-neglected aspect of the data-generating process has large effects on beliefs, consistent with the idea that incorrect mental representations are partly driven by selective attention.¹¹

A takeaway from this literature is that computations and the formation of mental representations interact: When computational complexity is high, subjects appear more likely to default to simplified representations that omit key features of the data-generating process. In other words, when decision makers face non-trivial uncertainty about both the right model and about aggregation, they frequently resolve the model uncertainty by jumping to one (often incorrect) model.

A second recurring result in this literature is low meta-cognitive awareness: confidence is

¹¹Interventions can improve representations by redirecting attention to neglected aspects. However, they may also create negative spillovers. Altmann et al. (2022) and Conlon (2025) both document that when people’s attention is exogenously directed to one previously-neglected part of the problem, their performance on this particular component increases, yet at the cost of lower performance in other parts of the problem.

at best weakly related to having the correct model (e.g., Enke, 2020; Graeber, 2023; Musolff and Zimmermann, 2025). This contrasts with cognitive-uncertainty work summarized in Section 4, where awareness of mistakes is more common. A key distinction appears to be between mistakes about the model or problem-solving approach (which people seem to have little awareness of) and mistakes in optimizing within a model (which people do seem to have awareness of), also see Enke et al. (2023).

Ignoring model uncertainty. A prominent idea in the recent literature is that when multiple models are possible, people do not average across models but, rather, focus on a single specific one and attempt to optimize within this mental framework, consistent with the evidence on system neglect.

Musolff and Zimmermann (2025) experimentally study mental models when the experimenter induces uncertainty about which model is the right one (i.e., even a fully rational subject would not be able to infer the correct model based on the information provided). They manipulate the computational complexity of translating a mental model into decisions and find that when computational complexity is high, people’s decisions appear to be based on the model that is most likely given the data and entirely neglect the other one (see Aina and Schneider, 2025, for a related result).

Augenblick et al. (2025a) present an experiment in which subjects make predictions about a stochastic process that involves both across-model uncertainty (how steep is the trend?) and within-model uncertainty (given a trend, how noisy are the data?). They also find that subjects’ predictions appear to largely ignore the across-model uncertainty, with subjects behaving as if they assume a single trend line and assess the degree of uncertainty only within this model, as if they cannot simultaneously entertain multiple models.

Neglecting statistics. The general idea that people simplify by entirely ignoring certain problem aspects is also present in experiments on belief formation that study how people combine base rates and likelihoods. A recurring pattern in the literature is that, in these problems, reported posterior beliefs often exhibit a multi-modal structure and cluster on a single statistic (Fan et al., 2023; Esponda et al., 2024). This multi-modality is very similar to the work on system neglect summarized above, and suggests that some people entirely ignore a problem feature that others take into account. For instance, Bordalo et al. (2026) show that subjects often rely on a single statistic, with selection driven by salience (how strongly a statistic favors one hypothesis). The authors measure attention allocation using self reports and find that subjects disproportionately attend to the salient statistic.

Related, Esponda et al. (2024) show the importance of selective attention for incorrect mental models through its effect on learning from feedback. They study a set-up of base rate neglect, in which experimental subjects know both the base rate and the diagnosticity of the signal. In addition, they sequentially observe feedback in the form of realizations from the

data-generating process. Many subjects decide to ignore the feedback by locking in their decision, thus getting locked into their incorrect mental model due to selective attention.

In the field, Hanna et al. (2014) study the behavior of seaweed farmers who need to take into account many input dimensions. These farmers entirely neglect the relevance of a particularly important input feature even though data on it is readily available. Consistent with an incorrect mental model driving a failure to optimize, farmers' decisions substantially improve when their attention is drawn to previously-unnoticed relationships. Just like in the case of the lab experimental evidence, there is thus evidence that selective attention is a major driver of incorrect mental models.

Narrow bracketing. The evidence reviewed so far concerns mental models for a single decision. Yet the logic of simplifying problems by entirely ignoring certain interrelationships also operates *across* decision problems: rather than solving two interrelated problems jointly, people solve them independently of each other. This 'compartmentalization' reflects an incorrect mental model because it pretends that decisions in one domain do not influence outcomes in other domains, see Read et al. (1999) for classic evidence.

Ellis and Freeman (2024) study the prevalence of bracketing strategies experimentally and develop tests that disentangle bracketing behavior from preferences. Using experiments in portfolio allocation, social allocation, and shopping, they find that approximately 74% of subjects are best described by (full) narrow bracketing. Thus, also in this context we observe that people *fully* neglect an interdependency, akin to the experiments on correlation neglect or model uncertainty discussed above. A key question is then *which* mental brackets people use. Imas (2016) reconsiders the puzzling evidence on whether risk taking increases or decreases in prior losses. By manipulating whether prior losses constitute paper losses or realized ones, he shows that people bracket *paper* losses and new investments together, while *realized* losses and new investments are mentally placed into separate brackets.

Neglecting contingencies. Many economic decisions require people to reason through future (unrealized) hypotheticals. Yet a rapidly-growing literature documents that people often mentally misrepresent decision problems through failures of contingent reasoning, by failing to properly attend to the decision-relevant states of the world, see Niederle and Vespa (2023) for a review. Esponda and Vespa (2016, 2023) document that the failure to properly condition on (and attend to) future hypotheticals is a driver behind multiple behavioral anomalies. They show that overbidding in auctions, mistakes in voting and the Ellsberg paradox are all partly driven by a failure of contingent reasoning. A main empirical innovation in this work is to switch the need to reason through multiple hypotheticals on and off, thereby directing people's attention to the main decision-relevant state of the world. For example, Esponda and Vespa (2023) document that the two-urn Ellsberg paradox is significantly weakened once people are prodded to focus on the decision-relevant state, in much

the same way as common-value voting improves once people are induced to focus on the state of being pivotal (Esponda and Vespa, 2014). Martínez-Marquina et al. (2019) show that people are substantially less likely to fall prey to the winner’s curse when they don’t need to reason through hypotheticals (also see Nagel et al., 2023).¹²

3.2 Analogies and Categorization: Learning from Similarity

Organizing equation and empirical tools. All of the approaches reviewed so far rest on the idea that people actually attempt to solve a decision problem by integrating the relevant problem aspects with the perceived model of the world. Sometimes, however, the decision maker’s uncertainty about how to interpret the available information (or his uncertainty about the true model) may be sufficiently large that he solves the problem by analogy. Work on analogies posits that people assess outcomes by retrieving the model from similar problems encountered in the past,¹³

$$\hat{m}(\cdot) = m^s(\cdot), \quad \text{where} \quad d^s \in \arg\max_{d^k} S(d, d^k). \quad (6)$$

Here, $S(d, d^k)$ is a similarity function between the current decision problem d and decision problems encountered in the past, d^k . The idea is thus that people solve problems by lumping them together into analogy classes or categories. This line of work builds on earlier research on attribute substitution (e.g., Kahneman et al., 2002). Research on this topic is closely related to work on similarity-based recall, except that here the focus is not on which *information* is remembered but on which *decision problems* are similar.¹⁴

Evidence. Salant et al. (2025) provide evidence on analogical reasoning in chess players. They study a version of chess in which the initial game configuration is random. They find that players are substantially more likely to use moves that they played in the past in games that are similar to the current random configuration. This analogical reasoning is more pronounced in less experienced players, suggesting that analogical reasoning is a simplification strategy when it is difficult to determine the optimal move.

In the lab, Fan et al. (2023) use similarity-based reasoning by analogy to shed light on the inference-forecast gap. This phenomenon says that people overreact more to recent information when they forecast a future realization of a stochastic process than when they

¹²Other applications of this logic include annuity take-up, public goods provision and redistributive behavior (Charness and Levin, 2009; Luttmer et al., 2023; Andre, 2025; Kendall and Chakraborty, 2022).

¹³See, for example, Bordalo et al. (2025c) and Gilboa and Schmeidler (1995) for similarity-based formulations and Mullainathan (2002b); Barberis and Shleifer (2003); Mullainathan et al. (2008); Jehiel (2005) for formulations without a similarity metric.

¹⁴A related body of work documents that people judge the quality of information according to similarity-based principles. Malmendier et al. (2026) show that people are more strongly influenced by information from people who are similar to them. Alsan et al. (2024) show that Black patients and their doctors are more likely to incorporate information from clinical trials that included a larger number of patients of color.

infer the underlying state. Fan et al. (2023) show that this is largely driven by the usage of similarity-based attribute substitution. They design experiments in which the statistics available to subjects to make forecasts or inferences differ in their similarity to the object subjects need to estimate. For instance, in a forecasting problem the target statistic is the expected outcome conditional on the realized signal, which is similar to the expected outcome conditional on the representative state. The authors manipulate similarity through contextual associations (labels) and find that subjects substitute the (difficult) forecasting or inference question they are asked for a (simple) statistic that is superficially similar.

In financial markets, Charles (2025) shows that categorization-based trading patterns affect market prices. The paper leverages the idea that companies may be lumped together in memory when they have overlapping earnings announcement schedules. The paper shows that when the announcements of two firms overlap, the separate earnings announcement of one of the firms several months later produces buying pressure in the other firm's stock. The interpretation is that because the two firms are associated in memory, the announcement of one firm directs attention also to the other one even when it does not announce earnings—they are placed in the same category, based on similarity.

3.3 Measurement of Mental Model Adoption

Most of the work summarized above attempts to infer people's mental models from their decisions and beliefs. A smaller-but-growing literature directly measures people's mental models, both in the lab (e.g., Fréchette et al., 2024) and in the field.

Lab evidence. A stream of recent papers studies which mental model people adopt when they observe specific suggestions of mental models, e.g. from an advisor. This is different from the studies summarized above, in which the decision maker only observed data rather than a suggested model. In these experiments, subjects are exposed to data, facing uncertainty about the causal structure that generates it. Subjects then observe messages about which causal structure could have generated the data, and which action to take as a result. Importantly, in most of these settings, if people could extract all relevant information from the data that is presented to them, the narratives they are exposed to would be redundant. Yet in practice, the suggested models strongly shape beliefs, suggesting that people indeed exhibit substantial model uncertainty (Barron and Fries, 2023; Kendall and Charles, 2022). This literature shows that people are most likely to adopt models that exhibit good fit of the data (Barron and Fries, 2023), exhibit a chain structure (Kendall and Charles, 2022), and are complex and / or make attractive promises (Ambuehl and Thysen, 2024).

Applied work. Applied research on the topic has often focused on the elicitation of mental models in the form of causal networks (or directed acyclic graphs, see Spiegler, 2016). Un-

like the lab experimental work summarized above, this body of research typically relies on more naturalistic elicitations, including open-ended survey questions. Two empirical results appear repeatedly. First, the mental models of households and managers are often fairly simple, and simpler than those of academic experts. Second, heterogeneity in mental models predicts heterogeneity in expectations.

Andre et al. (2023) study mental models of the stock market. They deploy open-ended survey questions asking both members of the general population and experts to explain how expected returns change in response to stale news. Academic experts mostly express models in line with market efficiency. Households and financial advisors, on the other hand, articulate mental models that are in line with ‘expected earnings reasoning’—directly equating higher expected earnings with higher expected returns. Heterogeneity in mental models is strongly predictive of variation in return expectations.

Using a similar methodology, Andre et al. (2026) study mental models of the macro economy. Groups of households and academic experts respond to open-ended survey questions about why they think inflation increased going from 2021 to 2022. The authors then represent the written responses as directed acyclic graphs. For example, the pandemic may have caused price gouging, which in turn caused inflation to increase. Andre et al. (2026) find that the mental models of households are considerably simpler than those of experts. Again, heterogeneity in mental models predicts expectations, here about future inflation.

Using similar techniques, Han et al. (2024) study the mental models of gas station managers. They ask managers what they believe contributes most to high profits. The main pattern is that managers with (separately measured) high cognitive skills appear to have fundamentally different mental models of their competitors. For instance, when managers are asked to predict their competitors’ response to a price cut, high-ability managers are more likely to predict a retaliatory price cut. These mental models are again correlated with economic decisions: managers who believe that competitors will retaliate for price cuts set significantly higher prices.

4 Aggregating Information Into Decisions

Work on the aggregation of information into decisions typically emphasizes that people rely on cognitive strategies that reduce the need to engage in a precise determination of value. Here, a main challenge is that carriers of value are typically disaggregated – they consist of multiple components that jointly determine overall value, requiring a cognitive process of integrating and trading off different problem dimensions. For example, labor supply involves trading off money and leisure; belief updating requires trading off a prior and a signal that may point in different directions; choice under risk calls for trading off risk and expected return; and so on.

4.1 Decision Uncertainty

The difficult process of aggregating the different components of an option into an overall evaluation or choice can give rise to uncertainty about what the best decision is. For instance, I am uncertain about which equity share (fraction of liquid wealth invested in stocks) maximizes my expected discounted utility. Such decision uncertainty can be measured both directly and more indirectly. First, some researchers ask experimental or survey participants unincentivized self-report questions about their certainty they made the best decision for themselves (e.g., Butler and Loomes, 1988, 2007; Cubitt et al., 2015; Enke and Graeber, 2023; Yang, 2023; Agranov et al., 2025). While sometimes these measures don't have a quantitative interpretation, in other cases researchers use quantitative elicitation formats that ask people how certain they are that their decision was actually their best decision ('cognitive uncertainty'). The dominant finding in this literature is that people routinely express uncertainty. For instance, in a study involving more than 30 experiments, Enke et al. (2024a) find that the vast majority of decisions is associated with cognitive uncertainty.

A second measure of choice uncertainty is choice noise. While the presence of decision noise has long been studied, recent research has shown that random choices are sometimes even made deliberately, as if people are revealing to the researcher that they do not really know what to choose. In experiments, subjects are asked to make the same choice multiple times on the same decision screen, or they are asked to choose a randomization probability for each option. Subjects oftentimes randomize, in preferential choice problems that potentially give rise to imprecise preferences (Agranov and Ortoleva, 2017, 2025; Agranov et al., 2023; Cettolin and Riedl, 2019), in objective decision problems in which imprecise beliefs could be present (Halevy et al., 2023), and in field decisions (Dwenger et al., 2018). Cognitive uncertainty and deliberate randomization are highly correlated empirically (Arts et al., 2024), suggesting that they capture similar underlying principles.

Third, people have been shown to delegate their decisions to a computer that estimates their preferences (Nielsen and Rigotti, 2022), or to abstract axioms that they endorsed earlier (Nielsen and Rehbeck, 2022), consistent with awareness of information-processing constraints. My interpretation is that these measures—cognitive uncertainty, deliberate randomization and delegating decisions—all capture people's uncertainty about how to aggregate problem primitives into a decision, and that this uncertainty reflects (i) cognitive limitations and problem difficulty and / or (ii) preference uncertainty..

Building on this evidence on the presence of decision uncertainty, empirical research often takes two forms. First, how does uncertainty predict decisions (e.g., attenuation and caution)? Second, how do people learn from environmental cues to reduce their decision uncertainty (e.g., comparison points and normalizations)? I discuss these in turn.

4.2 Responses to Aggregation and Tradeoff Uncertainty

4.2.1 Behavioral Attenuation

Organizing equation and empirical tools. Denote the degree of difficulty of aggregation (or the resulting decision uncertainty) by σ_i . Then, a first empirical regularity—behavioral attenuation—can heuristically be represented as¹⁵

$$\hat{v}_i(y_i^a) = \lambda_i(\sigma_i) v_i(y_i^a) + [1 - \lambda_i(\sigma_i)] d_i, \quad (7)$$

where $\lambda_i(\sigma_i)$ is a decreasing function and d_i a cognitive default. According to this equation, utility evaluations (or, in other formulations, observed decisions) are given by a convex combination of the truth and a fixed default, with the default weight increasing in decision uncertainty. The implication is that evaluations or decisions are insufficiently sensitive to variation in primitives ($\lambda < 1$) and that this inelasticity increases in uncertainty or problem difficulty (σ_i). In practice, attenuation will often boil down to a ‘pull-to-center’ effect.

Empirically, researchers have used three empirical strategies to document attenuation effects. First, several papers have shown that proxies for decision uncertainty—including cognitive uncertainty, dominance violations, across-trial inconsistencies in repetitions of the same problem, or objective mistakes in ‘mirror’ problems—are correlated with less sensitive decisions. Second, researchers have considered different exogenous variations that plausibly increase choice difficulty and uncertainty, including complexity and experience. Finally, researchers have relied on ‘flipping’ manipulations in which the researcher exchanges the decision variable and the fundamental: if the decision $a(x)$ is suspiciously insensitive, then difficulty-linked attenuation is indicated if in the inverse elicitation $x(a)$ is also insensitive because it implies that $a(x)$ is very sensitive (e.g., Shubatt and Yang, 2024; Wu, 2026).

Relevance. Attenuation may be of particular interest to behavioral economists from the perspective of tracing seemingly-distinct empirical regularities to a common mechanism because quite a few classic behavioral economics regularities boil down to an insensitivity of decisions to variation in a fundamental. Consider Figure 2, which displays three prominent behavioral economics regularities: the probability weighting function in choice under risk (Kahneman and Tversky, 1979), the hyperbolic discounting function in intertemporal choices over money (Loewenstein and Prelec, 1992; Cohen et al., 2020), and the attenuation puzzle in household equity shares (Giglio et al., 2021). Additional similar regularities include scope insensitivity in contingent valuation (Diamond and Hausman, 1994), economic expectations that appear to overreact for transient processes but underreact for persistent ones (Bordalo et al., 2020a; Bastianello and Imas, 2026), and the hard-easy effect in confi-

¹⁵See, for example, Gabaix (2019), Woodford (2020), Khaw et al. (2021), Frydman and Jin (2022) and Enke and Graeber (2023).

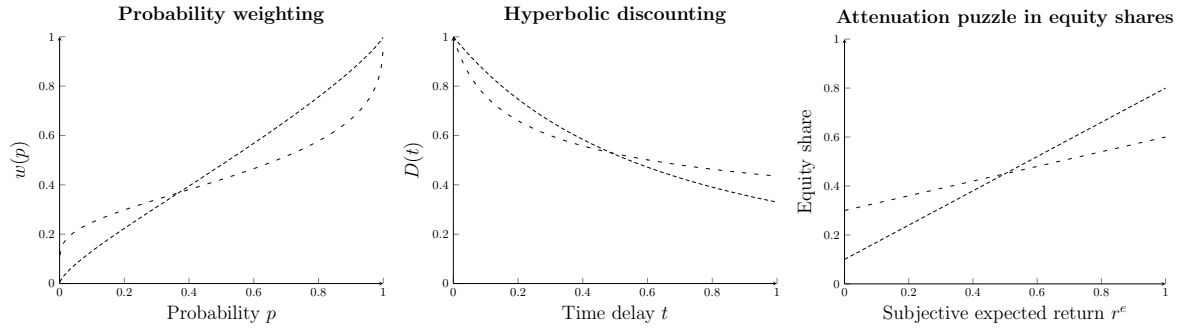


Figure 2: Examples of prominent behavioral economics regularities that reflect insensitivity

dence judgments (Moore and Healy, 2008). The key observation is that these regularities reflect a type of insensitivity, especially away from the boundaries of the parameter space.

The hypothesis behind research on behavioral attenuation is that these classic regularities—and others—reflect a common mechanism: the difficulty of aggregating the components of a problem into one’s best decision. Intuitively, if you’re uncertain how exactly a parameter such as a wage should influence your optimal decision, you may only partially respond to it because you also partly rely on what you typically do.

Evidence. Table 6 presents an overview of phenomena in the literature that can be understood as attenuation effects, emphasizing the wide range of methods different researchers have deployed to make related points.

Perhaps the most common application of the idea of attenuation so far is in the contexts of the probability weighting function and the hyperbolic discounting function over money. Both of these phenomena have been studied from various different methodological angles, delivering consistent results. First, probability weighting in experimental valuation tasks (or ‘pricing tasks’) is strongly correlated with proxies for decision uncertainty, including cognitive uncertainty (Enke and Graeber, 2023), mistakes in riskless ‘mirror’ problems (Oprea, 2024b) and dominance violations (Qiu et al., 2026). Second, probability weighting amplifies under experimental manipulations of uncertainty or difficulty, including in experience manipulations (Frydman and Jin, 2023) and manipulations that force subjects to sample from the outcome distribution (Oprea and Vieider, 2024). Finally, probability weighting ‘flips’ into ‘inverse probability weighting’ in the elicitation of probability equivalents rather than certainty equivalents (Bouchouicha et al., 2024; Imai et al., 2025; Wu, 2026; Shubatt and Yang, 2024), again consistent with decisions being attenuated functions of the exogenous problem fundamentals.

Similarly, hyperbolic discounting over money—often viewed as a puzzle given the fungibility of money—is likewise strongly correlated with cognitive uncertainty and choice noise (Enke et al., 2025), increases under complexity manipulations (Ebert and Prelec, 2007) and flips to ‘inverse hyperbolic discounting’ in the elicitation of time equivalents rather than money equivalents (Shubatt and Yang, 2024). More generally, time discounting is strongly

susceptible to cognitive manipulations such as enforced waiting times (Imas et al., 2022).

Moving beyond these two ‘classics’, Enke et al. (2024a) implement a large number of experiments that range from preference elicitations to belief updating to generic optimization problems, involving risk, time, consumption-savings, effort supply, taxes, fairness, prediction and inference, in both individual decisions and strategic games. They find that in nearly all tasks those decisions that are associated with higher cognitive uncertainty are more attenuated functions of the relevant economic primitive, suggesting that attenuation is a broad phenomenon that underlies various different empirical regularities, both old and new.

Evidence from the field. An emerging body of evidence points to the importance of behavioral attenuation also for field decisions. Drerup et al. (2017) and Giglio et al. (2021) study the attenuation puzzle in stock market participation: the puzzle that the link between future return expectations and equity share is quantitatively too small relative to canonical models. Both papers combine administrative data on stock holdings with survey data on expectations to show that the attenuation puzzle is strongly concentrated in people who exhibit high uncertainty about their beliefs (also see Charles et al. (2024) and Yang (2023)).

Augenblick et al. (2025b) study the role of behavioral attenuation for market-implied beliefs as reflected in option prices for the S&P500. They study whether prices are attenuated to the precision of information. When an option is far from expiration, the information in the market is weak and when it is close to expiration, the signals are very informative. Attenuation predicts that option prices move too much when the information is weak (far from expiration) yet too little when the information is strong (close to expiration). Augenblick et al. (2025b) document that this is the case, reconciling the joint existence of over- and underreaction of expectations as an outcome of attenuation to signal precision.

Salant and Spenkuch (2025) provide evidence on the role of complexity and attenuation in professional chess games. Using standard chess concepts, they quantify the complexity of each potential move and then study how the probability of choosing a move depends on its value and complexity. They find that the probability of choosing a winning move decreases in the option’s complexity, while the probability of choosing a losing move increases in the move’s complexity. As a result, the responsiveness of the player’s decision to variation in move quality decreases in complexity (i.e., complexity-driven attenuation).

Handel et al. (2026) identify cognitive-load driven attenuation effects in the types of diagnostic tests ordered by emergency department physicians. Leveraging within-physician variation, they show that when a doctor is under higher cognitive load (induced by more difficult previous cases), they regress more strongly to ordering common tests (what we can think of as ‘default’ tests), and thus exhibit attenuation to the patient’s actual condition.

Table 6: Empirical attenuation patterns across domains

Decision	Attenuation wrt	Concept in literature	Evidence	Examples
<u>Choice contexts in the lab</u>				
Certainty equiv. for lottery	Payout probability	Probability weighting	Insensitivity correlated w/ CU ; amplifies under complexity manip.	Enke and Graeber (2023)
Certainty equiv. for lottery	Payout probability	Probability weighting	Insensitivity correlated with insensitivity in obj. mirror problem	Oprea (2024b)
Certainty equiv. for lottery	Payout probability	Probability weighting	Insensitivity correlated w/ dominance violations	Qiu et al. (2026)
Certainty equiv. for lottery	Payout probability	Probability weighting	Turn p-weighting into inverse p-weighting by flipping elicitation mode	Bouchouicha et al. (2024); Wu (2026)
Lottery choice	Payout probability	Probability weighting	Forcing sampling from outcome distribution reduces insensitivity	Oprea and Vielder (2024)
Lottery choice	Payment	Value function curvature	Experience reduces insensitivity	Frydman and Jin (2023)
Choice under ambiguity	Likel. of ambig. event	Ambiguity insensitivity	Insensitivity correlated with magnitude of perceived ambiguity	Henkel (2022)
Portfolio allocation	Subj. expected return	Attenuation puzzle	Insensitivity correlated w/ CU ; amplifies under complexity manip.	Charles et al. (2024); Yang (2023)
Time discounting	Time delay	Hyperbolic discounting	Insensitivity correlated w/ CU / choice noise / obj. mirror problem	Enke et al. (2025)
Time discounting	Time delay	Hyperbolic discounting	Insensitivity increases under complexity manip.	Ebert and Prelec (2007)
Time discounting	Time delay	Hyperbolic discounting	Turn hyperbolicity into inverse hyperbolicity by flipping elicitation	Shubatt and Yang (2024)
Valuation of policy	Scope of policy	Scope insensitivity	Insensitivity correlated w/ CU ; decreases in flipped elicitation	Wu (2026)
Labor supply	Tax rate	Complexity manip.	Attenuation to tax rate	Abeler and Jäger (2015)
Labor supply	Wage	Attenuation to wage	Insensitivity correlated with CU	Enke et al. (2024a)
Belief reports	True beliefs	n/a	Strategic shading of beliefs towards 50-50	Danz et al. (2022)
<u>Beliefs in the lab</u>				
Belief updating	Signal strength	Under- and overreaction	Insensitivity correlated w/ CU	Enke and Graeber (2023); Augenblick et al. (2025b)
Information demand	Signal strength	Compression effect	Insensitivity correlated w/ CU ; amplifies under complexity manip.	Liang (2023); Enke et al. (2024a)
<u>Strategic games in the lab</u>				
Investment coordination	Value of project	Insensitivity to payoffs	Experience reduces insensitivity	Frydman and Nunnari (2023)
Beauty contest	Multiplier	n/a	Insensitivity correlated w/ CU	Enke et al. (2024a)
<u>Field experiments</u>				
Contingent valuation	Effectiveness of policy	Scope insensitivity	Insensitivity correlated w/ CU ; decreases with decision aids	Toma and Bell (2022)
Perception of tax rate	Tax rate	Schmedding	Perceived tax rate is attenuated function of true tax rate	Rees-Jones and Taubinsky (2020)
<u>Large-scale surveys on beliefs and perceptions</u>				
Stock market expectations	Historical probabilities	Pull-to-center	Insensitivity correlated w/ CU	Enke and Graeber (2023)
<u>Observational field data</u>				
Equity share	Subjective returns	Attenuation puzzle	Insensitivity correlated with low confidence	Drerup et al. (2017); Giglio et al. (2021)
S&P500 option price movement	Time to expiration	n/a	Price movement insensitive to informativeness of event	Augenblick et al. (2025b)
Sports betting beliefs	Time to end of game	n/a	Price movement insensitive to informativeness of event	Augenblick et al. (2025b)
Chess moves	Quality of move	n/a	Prob. of choosing move insensitive to quality when high complexity	Salant and Spenkuch (2025)
Physician's test orders	Patient condition	n/a	Cognitive load increases insensitivity of test to patient condition	Handel et al. (2026)

4.2.2 Caution and Complexity Aversion

Organizing equation and empirical tools. Caution or complexity aversion refer to a type of uncertainty aversion, i.e., the idea that valuations can heuristically be represented by¹⁶

$$\hat{v}_i(y_i^a) = v_i(y_i^a) - c_i(\sigma_i^a), \quad (8)$$

where $c(\cdot)$ is a ‘cost’ function that increases in evaluative uncertainty about an action or option, σ_i^a . This uncertainty aversion is reminiscent of ambiguity or risk aversion, except that it also arises in contexts without objective risk or ambiguity, i.e., when the only source of uncertainty is evaluative in nature and driven by the difficulty of aggregating known primitives into a decision.

This line of work has largely relied on two empirical techniques. First, to show that people undervalue options when they are more complex. Second, to show that these effects are more pronounced when cognitive uncertainty is higher.

Evidence. de Clippel et al. (2024) implement a series of experiments to document that valuations of complex bets are lower than their objective value, and that this undervaluation is strongly correlated with cognitive uncertainty. Their design is similar in spirit to an Ellsberg-type ambiguity aversion design, except that complexity (or cognitive uncertainty) replace ambiguity. For instance, in one experiment, subjects encounter a structured two-state belief updating problem in which many subjects are known to struggle with computing the Bayesian posterior. Subjects state their WTP for monetary bets on each state. In addition, they state their WTP for a known 50-50 bet. The idea behind the design is that, regardless of what the subjects’ beliefs in the updating problem are, at least one of the two WTP they indicate for the complex bet should be weakly larger than the WTP for the simple 50-50 bet. Instead, de Clippel et al. (2024) find that the WTP for both complex bets is much smaller. Moreover, they show that this reduction in WTP is strongly concentrated in subjects who indicate higher cognitive uncertainty about their beliefs in the updating problem, in line with the idea that cautious behavior reflects a type of ‘cognitive uncertainty aversion’.

This evidence is closely related to work on compound lottery aversion. Halevy (2007) and Gillen et al. (2019) show that subjects undervalue compound lotteries relative to the equivalent reduced lotteries in the same way that they undervalue ambiguous lotteries. In line with this, de Clippel et al. (2024) show that the degree of undervaluation of compound lotteries strongly increases in subjects’ cognitive uncertainty.

A related body of work has studied whether people are complexity averse with respect to the number of unique payout states of financial assets. There is evidence to suggest that financial assets with many distinct payout states are systematically undervalued or avoided (Huck and Weizsäcker, 1999; Bernheim and Sprenger, 2020; Puri, 2025). For example, Puri

¹⁶See, for example, Puri (2025) and Cerreia-Vioglio et al. (2024).

(2025) documents that subjects' valuations for risky lotteries—elicited through a multiple price list—strongly decrease in the number of payout states. Iyengar and Kamenica (2010) document that a preference for simpler lotteries is especially pronounced when the choice set contains a larger number of options.

Related, Fehr and Wu (2023) show that people shy away from products that are overly complex due to obfuscation of add-on features. This type of complexity aversion appears to be very similar to the aversion displayed against lotteries with many outcomes: when there are too many dimensions (or states) to consider, people are uncertain about the overall value of the object, and this leads them to undervalue or penalize the alternative.

4.3 Learning from Comparisons to Reduce Tradeoff Uncertainty

Many decisions may be non-trivial because they involve the need to navigate tradeoffs across different dimensions or problem components. This implicitly requires the decision maker to bring different problems dimensions into a “common currency” of value. A literature going back to Tversky (1969) highlights that to reduce the difficulty and costliness of such information processing people rely on comparative thinking: within-dimension comparisons rather than aggregation across dimensions. The simple intuition is that it is often cognitively difficult to produce an absolute assessment or valuation (e.g., how much is this wine worth to me?), yet relatively easy to produce a relative assessment (e.g., this wine is better than the one I had yesterday). Viewed through the lens of information processing emphasized in this review, one can thus view comparison points as information (Natenzon, 2019; Villas-Boas, 2024; Dean et al., 2026; Graeber & Enke, 2026).

In Section 2.2, I discussed the idea that comparisons may help people to infer which dimensions are important to pay attention to. Here, I discuss the complementary idea that comparisons also help with reducing tradeoff uncertainty because, for example, they help with interpreting scales and units, or with translating problem features into decisions.

4.3.1 Uncertainty and Comparative Reasoning

Organizing equation and empirical tools. A first line of work documents that people are more likely to rely on comparisons or reference points when the problem is more difficult or people exhibit greater decision uncertainty. This literature can heuristically be represented by positing that utility evaluations are given by¹⁷

$$\hat{v}_i(y_i^a) = v_i(y_i^a) + w(\sigma_i) [v_i(y_i^a) \pm v_i(r_i)], \quad (9)$$

where $w(\sigma_i)$ is an increasing function and r_i a reference or comparison quantity. The key comparative statics prediction is that evaluations (or decisions) are more strongly influenced

¹⁷See, for example, Villas-Boas (2024) and Dean et al. (2026).

by the comparison quantity when σ_i (complexity or decision uncertainty) is larger.

To test this idea, empirical researchers have generally manipulated decision uncertainty through experience and complexity variations. One relevant dividing line in this literature is whether the comparison quantity r is another choice option or a reference point.

Evidence. Focusing first on studies in which r is a choosable option, Arieli et al. (2011) analyze the effect of complexity on comparative reasoning by deploying eye-tracking techniques in a binary lottery choice context. They find that a large fraction of eye movements suggests within-dimension comparisons (payouts with payouts and probabilities with probabilities) rather than across-dimension aggregation. Most relevant from the perspective of the difficulty of aggregating tradeoffs, they find that the frequency of within-dimension comparisons increases in the complexity of the lotteries (also see Rubinstein, 1988).

Second, Hsee (1996) provides evidence that people rely more strongly on comparisons with other options when the option is more difficult to evaluate. In so-called joint-versus-separate evaluation studies, he shows that when a product dimension is very difficult to evaluate (for example because the units are unfamiliar), people largely ignore it when they evaluate a single option in isolation. Yet once two options get evaluated simultaneously, the difficult-to-value dimension strongly impacts valuations. The intuition is that even when people don't understand the units, they can still reason in comparative terms.

A second stream of papers has shown that that people also rely more strongly on comparisons with reference points (rather than other choice options) when they face greater uncertainty about their best decision. Graeber & Enke (2026) document in experiments that both contrast and assimilation effects with respect to exogenously-induced reference points are less pronounced when experimental subjects have more experience or better information. For example, subjects' willingness-to-accept for work offers is substantially less strongly influenced by the work offers of other subjects (the reference point) when they have more experience and / or when they have more information about the work task.

In the field, Jin et al. (2024) document that physicians' treatment decisions for patients are more strongly autocorrelated when the current patient is of greater medical uncertainty and when the physician is less experienced or more fatigued, again consistent with the idea that people rely more strongly on comparisons when they have more uncertainty.

Related, Graeber et al. (2025) document the role of valuation uncertainty in the stock market effects of earnings surprises. A well-known pattern is that the market appears to respond in part not just to absolute earnings announcements but also to categorical information about whether the earnings surprise is positive, negative or zero (a form of comparative thinking or reference dependence). By leveraging financial metrics of the difficulty of valuing firms, Graeber et al. (2025) show that this comparison effect is more pronounced for firms about which analysts exhibit greater valuation uncertainty.

In summary, this entire body of work suggests that comparative thinking—within-dimension

comparisons with other options or reference points—is more pronounced when people exhibit greater decision uncertainty, intuitively because comparative evaluations can often provide information even when absolute evaluations are difficult.

The insight that comparative thinking partly reflects decision uncertainty is not only of interest from a basic science perspective but also helps with predictions. A puzzle in empirical work is that comparison points sometimes push evaluations down (‘contrast effect’) and sometimes up (‘assimilation effect’). For example, labor supply increases in the reference point of typical daily earnings (an assimilation effect), yet decreases in the reference point of a typical hourly wage (a contrast effect). Graeber & Enke (2026) document that one can empirically predict the sign of comparison effects (contrast or assimilation) based on the idea that comparison points provide information about one’s best decision. Intuitively, reference points that consist of endogenous decision outcomes (such as typical daily earnings) provide information about the right ‘ballpark’ of the decision, producing assimilation. On the other hand, reference points that consist of exogenous problem attributes facilitates comparative reasoning (‘the current option is better / worse than the reference option’), producing a contrast effect. The authors provide evidence for this idea by manipulating whether comparison points consist of endogenous decision outcomes or exogenous attributes, consistently finding contrast effects when comparison points consist of exogenous attributes and assimilation effects when comparison points are given by endogenous decisions or outcomes.

4.3.2 Normalizations

Organizing equation and empirical tools. Normalizations are a particular type of comparative thinking effect according to which people implicitly ‘normalize’ all quantities by the average quantity or the range in the set. For example, the difference between 10 and 8 ‘looks’ larger when the third option is 11 than when it is 100. Informally speaking, the suggestion in the literature is thus that the perceived utility difference between options A and B is given by formulations of the type¹⁸

$$\hat{U}^A - \hat{U}^B = \sum \hat{v}_i(y_i^A) - \hat{v}_i(y_i^B) = \sum \frac{v_i(y_i^A) - v_i(y_i^B)}{N(y_i^A, y_i^B, \dots)},$$

where $N(\cdot)$ is a normalization function such as the average or range of (utility) outcomes in a dimension. Again, we can think of these normalizations as reflecting information about how to integrate different problem dimensions: when people don’t know how much a given problem dimension should influence their decisions (say, because they aren’t familiar with the units), then the range of outcomes or the average outcome in that dimension may provide information about how much the dimension is likely to matter. To take a trivial example, if I assess the value of one unit of an unknown foreign currency, it is clearly informative if

¹⁸See, for example, Bushong et al. (2021) and Webb et al. (2021).

the average price in that currency is one billion.

The key comparative statics prediction is that people implicitly weight dimensions more strongly the smaller the average or range is in that dimension.

Evidence. Several recent papers have provided evidence in line with such normalization effects. For instance, Somerville (2022) shows that in a binary choice between an expensive high-quality iPad and a cheaper low-quality iPad, people are more likely to choose the more expensive version when the range of prices is experimentally increased through a type of phantom decoy, consistent with the idea of people normalizing prices by the average price or the range of prices. In a related study, Dertwinkel-Kalt et al. (2017) document that consumers' propensity to purchase an expensive high-quality item increases in the expected price level, consistent with the idea that actual prices get normalized by expected prices. Similar normalization effects have been documented in the domains of lottery choice (Soltani et al., 2012) and effort supply (Bushong et al., 2021).

These findings on normalization stand in some tension to the results on focusing discussed in Section 2. Recall that the idea behind focusing is to overweight problem dimensions that have a large amount of variation, while the intuition behind normalization is to underweight them. However, upon closer inspection, these two patterns coexist because they capture different aspects of information processing: information acquisition and information aggregation. From the perspective of information acquisition, the 'differences-attract-attention' view probably dominates in high-dimensional problems in which attention is a scarce resource. In contrast, the normalization logic may apply to smaller choice sets in which people can directly compare all relevant options and dimensions, and the main difficulty lies in their aggregation rather than the acquisition of information. Bondi et al. (2025) provide direct evidence on this. They implement experiments on effort choice in which subjects rank two options (working more for more money or working less for less money), and the range of options in both dimensions is manipulated through decoys. Their main result is that as the options to be evaluated become more complex (more high-dimensional), the focusing logic gains in importance, meaning that subjects weight a dimension (effort costs or money) more strongly the more variation exists in that dimension.

5 Complexity and Information-Processing Demands

The previous sections have highlighted the importance of acquiring, interpreting and aggregating information for the purpose of making decisions. Yet, trivially, some decisions impose greater information processing demands than others. I here discuss research on what makes problems complex, following different definitions of 'complexity.'

Representational complexity. Representational complexity captures the difficulty of *encoding* the relevant objects and their interrelationships. Operationally, it increases in the number of items that must be simultaneously represented (e.g., options, attributes, signals, causal nodes) and in the number of relations among them (e.g., correlations, edges in a causal graph). A broad body of evidence supports the idea that cardinality affects complexity and resulting information processing demands. For example, the prominent literature on consideration sets and choice overload documents the strong effects of the number of choice options in the set on decision quality (Caplin et al., 2011; Reutskaja et al., 2011; Santos et al., 2012; Abaluck and Adams-Prassl, 2021; de Lara and Dean, 2024; Carvalho and Silverman, 2024). Yet even holding fixed the number of options, many researchers have documented strong effects of cardinality, by varying the number of distinct payout states in lottery choice (e.g., Huck and Weizsäcker, 1999; Puri, 2025; Arrieta and Nielsen, 2023), the number of future dates across which a payment is disaggregated (Dertwinkel-Kalt et al., 2022), the number of marginal tax rates (Abeler and Jäger, 2015), the number of distinct states of the world in belief updating problems (Ba et al., 2025), the number of causal nodes in a directed acyclic graph (Kendall and Oprea, 2022), the number of nodes in a game tree (Salant and Spenkuch, 2025) or the number of components of a strategic disclosure report (Jin et al., 2022). The general takeaway from this line of work is that information processing is more difficult when the number of items (or causal relations among them) is larger.

Computational complexity. Computational complexity captures the difficulty of *mapping* a given representation into a good decision. While likely influenced by many factors, some of the recent literature has emphasized the idea that computational complexity is partly governed by whether decisions involve pronounced tradeoffs across dimensions, dates or payout states. Heuristically speaking, this literature can be interpreted as suggesting that the perceived utility difference between options A and B is given by¹⁹

$$\hat{U}^A - \hat{U}^B = \frac{\sum v_i(y_i^A) - v_i(y_i^B)}{\Delta(v(y^A), v(y^B))} + \epsilon,$$

where $\Delta(\cdot)$ is a dissimilarity function such as $\Delta(\cdot) = \sum |v_i(y_i^A) - v_i(y_i^B)|$. According to this logic, options are more difficult to rank when they are more dissimilar from each other dimension-by-dimension, as this implies bigger tradeoffs. Empirical studies in this literature typically propose a measure of the strength of tradeoffs and then show that it predicts quantities such as cognitive uncertainty, choice noise and mistakes.

Enke and Shubatt (2023) show that a measure of the strength of tradeoffs across payout states is strongly predictive of cognitive uncertainty, behavioral attenuation, and choice noise in binary lottery choice decisions (also see Erev et al., 2010). Shubatt and Yang (2024) generalize this insight by showing that the strength of tradeoffs across dimensions (or dates,

¹⁹See, for example, Natenzon (2019), He and Natenzon (2024) and Shubatt and Yang (2024).

or states) strongly predicts cognitive uncertainty, choice inconsistencies and preference reversals in three domains: risky choice, intertemporal choice, and multi-attribute choice. Zhu et al. (2025) show that tradeoff complexity strongly predicts cognitive uncertainty and response times in strategic 2x2 matrix games. Gabaix and Graeber (2023) show that an un-incentivized measure of perceived complexity increases in the strength of intertemporal tradeoffs. Both Shubatt and Yang (2024) and Mezzadri and Woodford (2025) show that tradeoff complexity strongly predicts mistake rates in calculation problems with objective solutions. The estimated effects of tradeoff complexity are typically very large, suggesting that it is a first-order determinant of information processing requirements.

A second type of computational complexity that is highlighted in the literature is the presence of non-linearities. First, tax schedules are more difficult to understand when they are non-linear, and people tend to linearize them (Rees-Jones and Taubinsky, 2020). Second, Bayes' rule in updating beliefs implies strong non-linearities in how the precision of information should translate into beliefs, and recent evidence suggests that more pronounced linearities lead to more mistakes (Agranov and Reshidi, 2024). Third, research on household finance shows that people suffer from exponential growth bias, treating exponential growth as if it was linear (Stango and Zinman, 2009). Thus, the common element is that non-linearities make decision problems harder, and people often respond by linearizing them.

Implementation complexity. The approaches laid out above study the complexity of decision problems based on *features of the problem*. Another approach is to study the complexity of the *procedures, rules or algorithms* people use to solve a problem. This implementation complexity (or procedural complexity) captures the difficulty of *executing* a chosen solution, rule, or plan reliably—remembering it, applying it, and tracking its state. A recurring result in this experimental literature is that procedural complexity increases in—broadly speaking—different types of cardinality.

Oprea (2020) studies the complexity of algorithms (or rules) by eliciting subjects' willingness to pay to not implement (exogenously-assigned) rules. The author finds that subjects have a higher WTP to avoid implementing rules that have more states and involve more transitions between states. Related, Banovetz and Oprea (2022) study the endogenous selection of rules and document similar patterns—rules are avoided more when they have more states and involve more transitions. Still related, Kendall and Oprea (2022) document that the complexity of forming a mental model based on correlational data increases in the number of inputs and outputs that the true model involves, yet again a notion of cardinality.

Guan and Oprea (2024) provide evidence that these notions of implementation complexity matter for strategy selection in repeated games. Experimental subjects pick from a set of pre-defined strategies. When subjects need to implement their selected strategy themselves (rather than have it be implemented by a computer), they select strategies that score substantially lower on implementation complexity.

6 Taking Stock and Looking Ahead

The central message of this review is that many behavioral regularities can be understood through constraints on information acquisition, interpretation, and aggregation. An optimistic implication is that progress on cognitive foundations does not merely add new mechanisms; it also offers a more unified understanding of behavioral phenomena and a path toward better predictions. I discuss these in turn.

The cognitive foundations of some classics. The cognitive turn has begun to make meaningful progress on clarifying the cognitive foundations of several canonical behavioral regularities. Table 7 provides a (necessarily incomplete) overview of some “classics,” the puzzles they raise for standard accounts, and the types of evidence that have linked them to specific information-processing mechanisms. A useful way to read the table is that the “Cognitive foundations” column proposes a candidate mechanism, while the “Evidence” column highlights empirical strategies that provide evidence for cognitive explanations.

Two themes stand out. First, several classics appear to share related cognitive drivers, offering hope that behavioral economics may be able to reduce the number of concepts required to describe the economic implications of psychological factors. Second, cognitive foundations help explain why many classics are strongly procedure- and context-dependent.

Looking ahead: non-choice data and AI for naturalistic and field studies. The cognitive turn was partly enabled by new experimental paradigms, and it also suggests a productive evolution of experimental design. Classical experimental minimalism—stripping away context to isolate a theoretical construct under transparent incentives and procedures—has been enormously valuable. A complementary approach is *controlled naturalism*: re-introducing selected contextual elements (language, narratives, visual cues, realistic choice architectures) while preserving experimental control. This is particularly important when behavior is context- and procedure-dependent: rather than treating such dependence as “confounds,” controlled naturalism treats it as objects of study in light of cognitive mechanisms.

Helpful in this regard may be that the empirical toolkit for studying cognition may expand dramatically. Past work has leveraged non-choice data to identify cognitive foundations and connect them to behavior (see Camerer, 2008). Examples include elicitations of decision procedures (e.g., Arrieta and Nielsen, 2023), cognitive uncertainty (e.g., Enke and Graeber, 2023), response times (e.g. Rubinstein, 2016; Alós-Ferrer et al., 2021; Card et al., 2024), and process measures of information acquisition (e.g., Johnson et al., 2002; Camerer et al., 2005; Wang et al., 2010; Brocas et al., 2014; Reutskaja et al., 2011; Bohren et al., 2024). As designs become more naturalistic, machine learning and modern AI methods can potentially help quantify intrinsically high-dimensional inputs—text, speech, and images—in ways that preserve comparability across studies (Asirvatham et al., 2026; Bordalo et al.,

Table 7: Evidence on cognitive foundations of some classics

Phenomenon	Puzzles not explained w/o 'cognition'	Cognitive foundations	Evidence	Example references
<u>Choice</u>				
Probability weighting I	Procedure invariance (elicitation formats); description-experience gap;	Noisy cognition / beh. attenuation wrt prob.;	<i>CU</i> and mirror correlations; complexity and experience manip.	Enke and Graeber (2023), Oprea (2024b), Oprea and Vieider (2024), Frydman and Jin (2023)
Probability weighting II	Payoff dependence	Salience of outliers	Attention measurement and manip.	Bordalo et al. (2012), Bohren et al. (2024)
Hyperbolic discounting	Hyperbolicity over money; procedure depend. hyperbolicity also when early date is in future; Hyperbolicity $\neq$ front-end delay effect	Noisy cognition / beh. attenuation wrt delay	<i>CU</i> and mirror correlations; complexity, cognitive load and time pressure manip.	Ebert (2001), Ebert and Prelec (2007), Enke et al. (2025), Shubatt and Yang (2024)
Scope insensitivity	Existence not well-accounted for	Beh. attenuation wrt scope	<i>CU</i> correlation, decision aid manip.	Toma and Bell (2022), Wu (2026)
Attenuation puzzle in equity shares	Existence not well-accounted for	Beh. attenuation wrt expected return	<i>CU</i> correlation, complexity manip., default measurement	Giglio et al. (2021), Charles et al. (2024), Yang (2023)
Preference reversals in choice vs. valuation	Not well-accounted for	Noisy cognition / comparison complexity	Manipulate ease of comparison	Slovic et al. (1990), Butler and Loomes (2007), Shubatt and Yang (2024)
Skewness preferences	Context dependence	Differences attract attention	State space and complexity manip.	Dertwinkel-Kalt and Köster (2020), Bondi et al. (2025)
Ellsberg paradox	n/a	Contingency neglect	Contingent reasoning manip.	Esponda and Vespa (2023)
Overbidding in auctions	Sequential vs. simultaneous decisions	Contingency neglect	Contingent reasoning manip.	Esponda and Vespa (2023), Martínez-Marquina et al. (2019) Nagel et al. (2023)
<u>Beliefs and expectations</u>				
Over- / underreaction to info I	Variation across contexts	Beh. atten. wrt signal strength;	Obj. benchmark; <i>CU</i> correlation	Augenblick et al. (2025b), Enke and Graeber (2023)
Over- / underreaction to info II	Variation across contexts	Overw. salient / similar features	Attention and similarity manip.	Bordalo et al. (2026), Fan et al. (2023), Ba et al. (2025)
Over- / underreaction to info III	Variation across contexts	Similarity-based recall	Memory cues	Gennaioli et al. (2024), Butera et al. (2024)
Stereotyping	Not well-accounted for	Selective memory	Selective retrieval of represent. info	Bordalo et al. (2021), Misserocchi (2023), Conlon and Patel (2022)
Full system neglect (correlation, selection etc.)	Context dependence	Incomplete representation	Attention and complexity manip.	Enke and Zimmermann (2019), Enke (2020) Graeber (2023), Musolf and Zimmermann (2025)

2025c; Graeber et al., 2023a,b), and algorithms that predict visual salience can help explain attention and resulting choices (Li and Camerer, 2022).

Finally, future work is likely to document cognitive foundations in increasingly ecological settings. A main challenge is measurement: cognitive concepts are often defined not only over payoffs, but also over primitives such as complexity, decision uncertainty and similarity that are rarely measured in the field thus far. One hope is that machine learning and AI will enable the increased measurement of these constructs. Only then will the cognitive turn be able to deliver on its goal of enabling better, context-dependent predictions.

7 References

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ONLINE APPENDIX

A Details on AI Classification of Papers

The sample consists of all articles published between 2000 and 2025 in top 5 journals (*American Economic Review*, *Econometrica*, the *Journal of Political Economy*, the *Quarterly Journal of Economics*, and the *Review of Economic Studies*). Bibliographic data was obtained via Clarivate’s Web of Science Core Collection. I restrict the sample to records with Document Type = “Article”, excluding proceedings papers, early-access entries, and retractions. Whenever possible, missing abstracts were filled in via API queries to Crossref and OpenAlex.

Each paper is classified using Anthropic’s Claude Opus 4.6 large language model in two sequential passes. In the first pass, the model receives each paper’s title and abstract—and is explicitly prohibited from using journal name, author identity, or outside knowledge—and returns four outputs: (i) a binary decision on whether the paper belongs to the behavioral economics literature; (ii) one or two topic categories from {Cognitive, Biases, Social, Risk & Time, Strategic, Gender, Other}, ordered by importance; (iii) zero to three research type labels from {Theory, Lab/Online Experiment, Observational Data}; and (iv) exactly one field label from {Abstract/Mechanism, Applied Micro, Finance, Macro}. In the second pass, another model instance is asked to confirm the first pass’ conclusions. Note that the upper panels in Figure 1 plot the most pertinent, primary category for each paper. The bottom right panel weights each design category by the inverse share of total design categories that a paper is assigned to. Further, the ‘gender’ category was merged into ‘other’ for visualization purposes. The full prompts are reproduced below.

System Prompt (First Pass)

You are an expert research assistant in economics. You MUST use ONLY the title and abstract provided (do NOT use journal, authors, reputation, or outside knowledge).

Output TWO sections in plain text:

Section 1 begins with “REASONING:” and explains the decision briefly.

Section 2 begins with “JSON:” and contains STRICT JSON only (no markdown fences).

Be conservative: assign labels ONLY if clearly supported by title/abstract; avoid over-inclusion.

User Prompt (First Pass)

TASK (do all three for EACH item):

Return one JSON object per item with:

- `include_behavioral` (boolean)

- `categories` (array; 0–2 of: COGNITIVE, SOCIAL, RISK_AND_TIME, BIASES, STRATEGIC, GENDER; or ['OTHER'] alone)
- `research_type_labels` (array; 0–3 of: THEORY, LAB_OR_ONLINE_EXPERIMENT, OBSERVATIONAL_DATA)
- `field_label` (string; one of: ABSTRACT_MECHANISM, APPLIED_MICRO, FINANCE, MACRO; use null if not behavioral)

Output format (strict JSON only):

```
{{"results":[{"key":"...", "include_behavioral":true, "categories":["COGNITIVE"], "field_label":"APPLIED_MICRO"}]}}
```

1) Behavioral Economics Gating + Topic Categories

Step A (gating). Decide if the paper is part of the behavioral economics literature. A paper counts as behavioral economics ONLY if a behavioral or psychological concept is a CORE part of the paper's analysis or motivation (not just mentioned in passing). A paper is NOT necessarily behavioral simply because people have limited/imperfect information. A paper is not behavioral simply because it studies beliefs, perceptions, or deviations from a theoretical benchmark; it is behavioral only if those deviations are explicitly modeled or interpreted as arising from psychological mechanisms.

If NOT behavioral: set `include_behavioral=false` and IMMEDIATELY set `categories=[]` and `field_label=null` (do not force-fit).

If behavioral: set `include_behavioral=true` and assign categories using rules below.

Before assigning `include_behavioral=true`, you MUST be able to answer YES to at least ONE of the following based ONLY on the title/abstract:

1. The abstract explicitly references a psychological mechanism (e.g., bias, cognitive limitation, heuristic, social preference, identity, present bias, loss aversion), AND it is central to the paper (not a passing mention).
2. The abstract explicitly tests for a deviation from a rational benchmark/model (e.g., systematic departures from rational expectations/standard theory/optimal choice), and treats that deviation as a main object of study (not a minor side result).

If you cannot answer YES to (1) or (2), you MUST set `include_behavioral=false` (even if the paper discusses beliefs, perceptions, information frictions, heterogeneity, or slow adjustment).

Anti-false-positive exclusion rule (override). Set `include_behavioral=false` if the abstract's main pattern can be explained as pass-through/adjustment/frictions/measurement relative to standard theory, AND

the abstract does NOT explicitly name a psychological mechanism or behavioral bias. Common non-behavioral phrases that are NOT sufficient on their own:

- perceived / beliefs / expectations / confidence / survey beliefs / disagreement / heterogeneity
- wedge / gap vs. standard theory / deviation from one-to-one mapping
- slow incorporation / sluggish adjustment / delayed pass-through / sticky rates
- frictions / information frictions / search frictions / costs / constraints
- inattention or salience used only as “economic frictions” without psychological interpretation

In these cases, you MUST output `include_behavioral=false` unless the abstract explicitly frames the mechanism as psychological (bias/heuristic/limited attention/etc.) or explicitly tests a behavioral deviation as such.

Category assignment rule (if behavioral only). Choose ONE or TWO categories among: COGNITIVE, SOCIAL, RISK_AND_TIME, BIASES, STRATEGIC, GENDER. Return TWO categories only if two clearly and centrally apply.

Important ordering rule: If you return TWO categories, the FIRST must be the PRIMARY (most central) category, and the SECOND must be the SECONDARY category. Do NOT order alphabetically; order by centrality/importance in the title/abstract. If exactly one applies: return exactly one. If none of the six apply BUT it is behavioral: return OTHER as the ONLY category. OTHER must NEVER appear alongside another category.

Category Definitions

COGNITIVE (attention, memory, salience, complexity, decision uncertainty, mental models). Assign if the mechanism or measured outcome involves a SPECIFIC cognitive channel, e.g., limited attention/inattention/salience; memory/forgetting; complexity; choice difficulty/decision uncertainty; reasoning by analogy/categorization; context dependence; incorrect mental models. Also assign if centrally about cognitive constraints in information processing/acquisition. Do NOT assign if cognition is only generic (e.g., “bounded rationality,” “heuristic,” “naive,” “mistake”) without a specific cognitive channel/measure/test. Do NOT assign simply because the setting has limited/imperfect information or learning.

SOCIAL (social preferences, norms, morality, moral values, identity, social image, self-image, giving, fairness). Assign if centrally studies altruism/prosociality/charity/giving; fairness/inequity aversion; reciprocity; moral

preferences/values; social norms and compliance; identity-driven behavior (social identity, self-image, social image) as main mechanism or main outcome. Do NOT assign for mere social context (e.g., “social networks,” “social welfare,” “social insurance”) unless explicitly about social preferences/norms/identity/morality.

RISK_AND_TIME (risk preferences, time preferences, present bias, probability weighting, loss aversion, reference dependence). Assign if centrally studies decision-making under risk/uncertainty or intertemporal choice: risk preference parameters; probability weighting; ambiguity aversion; loss aversion; reference dependence; Allais-type patterns; discounting/time preference; present bias/hyperbolic discounting. Do NOT assign if risk/time is only part of the setting (e.g., “risky assets”) without estimating/testing preference components or behavioral choice patterns.

BIASES (judgment/choice biases, fallacies, neglect, paradoxes, systematic errors). Assign if centrally studies systematic deviations from rational benchmarks framed as biases/errors: base-rate neglect, projection bias, overconfidence, confirmation bias, gambler’s fallacy, hot-hand fallacy, framing effects, limited consideration, belief distortions, paradoxes, neglect of information, systematic mistakes in inference. Do NOT assign if the paper is merely “behavioral”/“bounded rationality” without a specific bias/error pattern as core contribution. Do NOT assign if it tests a specific cognitive mechanism better categorized as COGNITIVE.

STRATEGIC (behavioral game theory: Level-k, cognitive hierarchy, quantal response, bounded strategic reasoning). Assign if centrally studies strategic interaction with behavioral strategic reasoning: Level-k, cognitive hierarchy, quantal response equilibrium, noisy best response, limited depth of reasoning, behavioral equilibrium selection, experimental game behavior explained via these models. Do NOT assign if the paper is about game theory/IO/auctions generally without explicitly modeling/testing behavioral strategic reasoning.

GENDER (gender differences, stereotypes, discrimination tied to gender). Assign if centrally studies gender differences in preferences/behavior, gender identity effects, gender stereotypes, or gender-based discrimination. Do NOT assign if gender is not a central explanatory variable.

2) Research Type Labels (0–3 may apply)

Goal: identify research approach/evidence type (not topic). Assign any that clearly apply; otherwise leave empty.

THEORY. Assign if core contribution is theoretical: formal model, equilibrium analysis, analytical propositions, proofs, welfare analysis derived from a model, calibrated/simulated model used primarily as a theoretical object. Signals: model, theory, proposition, lemma, proof, equilibrium, analytical, derivation, structural model

(if primarily model-based rather than estimating from microdata). Do NOT assign if mainly empirical estimation that only mentions a model as motivation.

LAB_OR_ONLINE_EXPERIMENT. Assign if primary evidence comes from experiments with randomized assignment/participant recruitment typical of lab/online/field experiments. Includes: lab, online (MTurk/Prolific), survey experiments, RCTs, field experiments, A/B tests, incentivized tasks, dictator/ultimatum/public goods games, randomized information treatments. Do NOT assign if it is only a natural experiment/quasi-experimental design on observational data (that is OBSERVATIONAL_DATA).

OBSERVATIONAL_DATA. Assign if uses non-experimental data where researcher does not randomize treatment: administrative, panel, non-experimental surveys, transaction/market data, medical records, naturally occurring text-as-data, archival datasets; or identification strategies like DiD, IV, RDD, event study, natural experiment, matching, structural estimation using observed data. Note: quasi-experimental strategy on existing data is still OBSERVATIONAL_DATA.

Overlap/edge rules. THEORY + (LAB_OR_ONLINE_EXPERIMENT and/or OBSERVATIONAL_DATA) is allowed if both develop a model and test/estimate it. If abstract is vague about evidence (e.g., “we study” without methods/data), return [] unless THEORY is clearly indicated. Surveys: “survey experiment” or randomized vignettes → LAB_OR_ONLINE_EXPERIMENT; descriptive/correlational survey analysis → OBSERVATIONAL_DATA.

3) Field Label (exactly one if behavioral; null otherwise)

Only assign field_label if include_behavioral=true. If include_behavioral=false, field_label MUST be null. Choose EXACTLY ONE of: ABSTRACT_MECHANISM, APPLIED_MICRO, FINANCE, MACRO.

ABSTRACT_MECHANISM. Assign if main contribution is a general mechanism/model/behavioral or decision-theoretic insight not primarily situated in a specific applied domain. Signals: general theory/model of decision-making; broad mechanism without clear domain; methodological/conceptual contribution; experiments aimed at identifying a mechanism rather than a domain-specific question. If clearly domain-focused (labor/public/dev/finance/macro), do NOT use this.

APPLIED_MICRO. Micro-applied real-world institutions/policies/markets/people: labor, public, development, education, health, discrimination, household behavior, firms in micro settings; IO/market design when clearly micro-applied.

FINANCE. Financial markets/institutions/assets: asset pricing/returns, trading, portfolio choice, corporate finance, banking/credit, intermediation, household finance, market microstructure, risk premia.

MACRO. Aggregate: business cycles, inflation, monetary/fiscal policy at macro level, growth, aggregate unemployment, DSGE/aggregate equilibrium, macro expectations, aggregate time series.

Tie-break rules (choose ONE): If clearly finance (assets/banks/credit markets as core), choose FINANCE even if micro methods. If clearly macro (aggregate policy/cycles/DSGE/aggregate time series), choose MACRO. If clearly labor/public/dev/education/health/policy evaluation, choose APPLIED_MICRO. Otherwise, choose ABSTRACT_MECHANISM.

Final Consistency Rules. If `include_behavioral=false`: categories MUST be [], `field_label` MUST be null. If `include_behavioral=true`: categories MUST have length 1 or 2; if category is OTHER it must be the only element. `research_type_labels` may be empty or include 1–3 labels; never invent methods not stated/implied by title/abstract.

Strictness: Apply the SAME gating strictness to every item regardless of how many items are in the list (one or many). Do not be more lenient in large batches; when in doubt on behavioral gating, set `include_behavioral=false`.

Output. REASONING: Provide per-item reasoning. Keep it short and keyed (e.g., “key=...: ...”). If two categories: explicitly state “Primary category=..., Secondary category=...”. JSON: Output the STRICT JSON object only. No extra characters before {.

[Followed by the batch of papers, each as a JSON object with key, title, and abstract.]

System Prompt (Second Pass / Audit)

You are a SKEPTICAL senior reviewer in behavioral economics. Your job is to RE-EVALUATE papers that were previously classified as behavioral economics and REJECT those that do not truly belong. Your DEFAULT is to REJECT. You should only CONFIRM a paper as behavioral if the evidence is clear and unambiguous. You MUST use ONLY the title and abstract provided.

Output TWO sections in plain text:

Section 1 begins with “REASONING:” — your per-item reasoning (keyed by paper key).

Section 2 begins with “JSON:” — STRICT JSON only (no markdown fences).

User Prompt (Second Pass / Audit)

You are re-evaluating papers that were PREVIOUSLY classified as behavioral economics. Your task: for each paper, decide whether the classification should be CON-

FIRMED or REJECTED.

Mandatory Checklist (answer ALL four questions for EACH paper):

Q1: Does the abstract explicitly NAME a specific psychological mechanism? (e.g., present bias, loss aversion, overconfidence, social preferences, limited attention, projection bias, reference dependence, probability weighting, reciprocity, identity, etc.)

Answer: YES — name it / NO

Q2: Is that psychological mechanism CENTRAL to the paper (a core part of the analysis, not a passing mention, background motivation, or minor extension)?

Answer: YES / NO / N/A (if Q1=NO)

Q3: Could the paper's main contribution be fully explained by standard economic theory (rational agents + frictions, information costs, search costs, standard preferences)?

Answer: YES (standard theory suffices) / NO (genuinely needs behavioral component)

Q4: Is the paper PRIMARILY about one of these NON-behavioral topics that often gets misclassified?

- Information economics (Bayesian learning, information acquisition, signal extraction)
- Beliefs/expectations WITHOUT a named psychological bias
- Heterogeneous agents or heterogeneous beliefs (without behavioral mechanism)
- Market frictions, search frictions, adjustment costs
- Incomplete contracts, mechanism design, or asymmetric information
- Uncertainty or ambiguity in a purely decision-theoretic (non-behavioral) framework

Answer: YES (misclassified — reject) / NO

Decision Rule: CONFIRM as behavioral ONLY IF: Q1=YES AND Q2=YES AND Q3=NO AND Q4=NO. REJECT (set `confirm_behavioral=false`) in ALL other cases.

Examples of Papers That Should Be Rejected

Reject Example 1:

Title: "Inflation Expectations and Consumer Spending"

Abstract: "We study how households' inflation expectations affect their spending decisions using survey data. We find that higher expected inflation leads to more current

spending...”

Q1=NO (no psychological mechanism named), Q2=N/A, Q3=YES, Q4=YES (beliefs without bias)

→ REJECT. Studies expectations but no behavioral/psychological mechanism.

Reject Example 2:

Title: “Information Frictions in the Housing Market”

Abstract: “We show that search frictions and information asymmetries cause price dispersion in housing markets. Buyers who search more intensively obtain better prices...”

Q1=NO, Q2=N/A, Q3=YES (search frictions are standard), Q4=YES (market frictions)

→ REJECT. Standard search/information frictions, no psychological mechanism.

Reject Example 3:

Title: “Disagreement Among Forecasters and Monetary Policy”

Abstract: “We document substantial disagreement among professional forecasters about future inflation and show this affects monetary policy transmission...”

Q1=NO, Q2=N/A, Q3=YES, Q4=YES (heterogeneous beliefs without behavioral mechanism)

→ REJECT. Disagreement/heterogeneous beliefs, no named psychological bias.

Examples of Papers That Should Be Confirmed

Confirm Example 1:

Title: “Present Bias and Credit Card Borrowing”

Abstract: “We show that consumers with present-biased preferences systematically over-borrow on credit cards. Using a field experiment, we find that present bias explains 40% of...”

Q1=YES (present bias), Q2=YES (central), Q3=NO (needs behavioral), Q4=NO

→ CONFIRM. Present bias is an explicit psychological mechanism, central to analysis.

Confirm Example 2:

Title: “Loss Aversion in Labor Supply”

Abstract: “We test whether taxi drivers exhibit reference-dependent preferences by examining daily labor supply decisions. Consistent with loss aversion around a daily income target...”

Q1=YES (loss aversion, reference dependence), Q2=YES, Q3=NO, Q4=NO

→ CONFIRM. Loss aversion / reference dependence explicitly named and central.

Output Format

REASONING:

key={key}: Q1=... Q2=... Q3=... Q4=... → CONFIRM/REJECT because ...

(repeat for each paper)

JSON:

```
{"results":[{"key":"...", "confirm_behavioral": true/false,  
  "rejection_reason": "...or null if confirmed"}]}
```

rejection_reason: If rejected, give a concise reason (e.g., “beliefs without psychological mechanism,” “standard information economics,” “frictions not behavioral”). If confirmed, set to null.

[Followed by the batch of papers, each as a JSON object with key, title, abstract, and original_categories from the first pass.]

B Additional Tables

Table 8: Evidence for complexity aversion and caution

Decision	Complexity / uncertainty	Evidence	Result	Example references
Complexity from aggregation, incl. pref. uncertainty				
WTP for lottery	Compound prob.	Complexity manip.	WTP ↓ when compound	Halevy (2007); Gillen et al. (2019)
WTP for bet	Compound prob.	Complexity manip.; <i>CU</i> correl.	WTP ↓ when compound + higher <i>CU</i>	de Clippel et al. (2024)
WTP for bet	Update beliefs	Complexity manip.; <i>CU</i> correl.	WTP ↓ when update beliefs + higher <i>CU</i>	de Clippel et al. (2024)
Prosociality	Pref. uncertainty	Uncertainty manip.; <i>CU</i> correl.	Giving ↓ when higher uncertainty	Chakraborty and Henkel (2025)
Complexity from cardinality				
Lottery choice	# of options	Complexity manip.	Choose safe when more options	Iyengar and Kamenica (2010)
Lottery choice	# of states	Complexity manip.	Avoid options with many states	Huck and Weizsäcker (1999)
WTP for lottery	# of states	Complexity manip.	WTP ↓ when more states	Bernheim and Sprenger (2020); Puri (2025)
Opt out	# of options	Complexity manip.	Opt out ↑ when more options	Carvalho and Silverman (2024)
Demand for product	# of add-ons	Complexity manip.	Demand ↓ when more add-ons	Fehr and Wu (2023)
Field evidence				
Asset purchase	# of outcomes	Binary option vs. dominating bull spread	Purchase dominated asset	Goodman and Puri (2025)